

# Monotonicity of the stochastic discount factor and expected option returns

Ranadeb Chaudhuri\*

and

Mark Schroder†

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## Abstract

Empirical evidence shows that the pricing kernel, or stochastic discount factor (SDF), is not always downward sloping when estimated using S&P 500 data. On the other hand, we show that individual stock SDFs are in general downward sloping. We show that a simple jump diffusion returns model can reconcile these empirical findings. The same model also implies a steeper implied volatility curve for the index compared to the typical stock, a well-known empirical fact from the options literature. Both the SDF results and the implied-smile results can be explained by a common source of jump risk among stocks together with diversification of Brownian risk in the index. SDF violations arise when the per-unit-return compensation for Brownian risk exceeds that for jump risk. Diversification increases the former compensation more than the latter, making SDF violations more likely in the index than the typical stock. We also devise novel empirical tests of overall SDF monotonicity based on average returns of option trading strategies, thus avoiding the estimation of the density functions of returns.

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\*Visiting Assistant Professor of Finance, Kelley School of Business, Indiana University, Bloomington, IN 47405. Tel.: (812) 855-9270, E-mail: [chaudhur@indiana.edu](mailto:chaudhur@indiana.edu).

†Department of Finance, The Eli Broad College of Business, Michigan State University, East Lansing, MI 48824-1121. Tel.: 517-432-0622, E-mail: [schroder@bus.msu.edu](mailto:schroder@bus.msu.edu). We are thankful for many helpful comments from Mark Loewenstein, Paolo Sodini, and seminar participants at Michigan State University, Oakland University, Nanyang Business School, Stockholm School of Economics, Auburn University, and the 2010 Western Finance Association Meetings. We are grateful to Stephanie and Valerie Schroder for editing the paper.

A large literature has been devoted to estimating the stochastic discount factor (SDF), mainly because the SDF projected on market returns reveals marginal utility in a simple representative agent model. Even outside the representative-agent setting, the shape of the SDF, whether projected on index or individual stock returns, contains useful information about the distribution of returns and risk premia. We show that in a jump diffusion model, a hump in the shape of the projected SDF (whether for a stock or index) indicates a concentration of the jump-component of the return near the hump, as well as a disparity in the per-unit-return compensations of jump and diffusion risk.

Previous work has almost exclusively dealt with projections of the SDF on indexes, typically testing whether it is a decreasing function of S&P 500 returns, as predicted by a simple one-factor representative agent model. The preponderance of the evidence suggests violations of monotonicity. Various explanations for violations have included nontraded income or state variables (see Brown and Jackwerth (2004)); state dependence in the representative agent's utility (Brown and Jackwerth (2004)); heterogeneous investor beliefs (Ziegler (2007), Bakshi and Madan (2009), and Hens and Reichlin (2010)); the influence of a latent state variable (Chabi-Yo, Garcia, and Renault (2008)); and investor sentiment (Han (2008)).

Contrary to the evidence on equity-index SDFs, our empirical evidence suggests that SDFs of individual stocks are in general monotonic.<sup>1</sup> The average stock SDF has a downward sloping projected SDF, in contrast to the index, which we find exhibits a hump on the left (corresponding to negative returns). We show that a simple jump diffusion model can explain SDF monotonicity for the typical stock yet monotonicity violations for the index.<sup>2</sup> The same model also implies a steeper (normalized) implied volatility curve for the index compared to the typical stock, a well-known empirical fact from the options literature (see, for example, Dennis and Mayhew (2002); Bakshi, Kapadia, and Madan (2003); and Duan

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<sup>1</sup>As defined in Section I, we say that the SDF is strictly monotonic if the projected SDF (the expected value of the SDF conditional on the period-end stock price) is strictly decreasing in the stock price.

<sup>2</sup>Though the simple jump diffusion model is not novel, a similar model is used by Branger and Schlag (2004) to explain the steepness of the index smile, our characterization of SDF monotonicity is.

and Wei (2009)).<sup>3</sup> If the idiosyncratic part of jump risk is small compared to the idiosyncratic part of diffusion risk (for example, stocks share a large common jump component), then diversification in the construction of the index has two effects: 1) it accentuates the relative jump risk of the index, thereby increasing the smile slope of the index relative to the smile slope of the typical stock, and 2) it increases the per-unit-return compensation of the Brownian risk relative to jump risk. The second effect tends to make SDF humps more likely in the index than the typical stock, but humps in our model can only arise if the per-unit-return compensation is strictly greater for the Brownian part of risk. SDF humps, therefore, are not necessarily implied by a steeper index smile.<sup>4</sup> Furthermore, the smile slope measures the (risk-neutral) expected jump part of return relative to diffusion risk, and is invariant to both the market price of Brownian risk and the concentration of jump risk. We nonetheless find empirically that the smile slope is a good indicator of SDF humps. A negatively (positively) sloped smile indicates downward (upward) jump potential, essentially a necessary condition for an SDF hump on the left (right) side of the graph.

Although our empirical results suggest that the average stock has a downward sloping projected SDF, violations are not uncommon. Our model predicts that stocks exhibiting stronger downside (upside) jump potential will more likely exhibit humps in the SDF corresponding to negative (positive) returns. We use a key insight by Yan (2011), that stocks with downside (upside) jump potential will exhibit negative (positive) at-the-money (ATM) implied-volatility slopes, and sort stocks according to the normalized ATM slope. Consistent with the theory we find that indeed SDFs tend to exhibit left-side humps for the stocks in the lowest quintile (which have a strong negative slope), and right-side humps for the top quintile (which has a positive average slope).

Another contribution of our paper is that we devise a test of SDF monotonicity based

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<sup>3</sup>Based Proposition 5 (d) below, we normalize by scaling the implied volatility slope by the at-the-money strike. Most of the papers showing a steeper index implied smile also divide by the implied volatility. See subsection C for more information.

<sup>4</sup>As discussed in Section IIA, in a simple representative-agent setting with aggregate wealth equal to the index value, the per-unit-return compensations for the different risks must match in equilibrium.

on option-trading strategy returns that is model-free and avoids the need to estimate the SDF. Furthermore, it is straightforward to apply statistical tests of overall monotonicity. Our approach, although applicable to all assets with traded options, is particularly useful for individual stocks because the (typically) limited number of available strikes prices compounds the challenges of applying the traditional estimation approaches. The typical empirical approach is to estimate both the risk-neutral density (from option prices), and the objective density (from historical stock returns), and take the ratio (see, for example Jackwerth (2000), Barone-Adesi and Dall'O (2010) and Barone-Adesi, Engle, and Mancini (2008)).<sup>5</sup> Chabi-Yo, Garcia, and Renault (2008) show that failure to condition on the full information set (i.e., latent state variables) can result in spurious nonmonotonicity. Moreover, because it is a ratio, unbiased conditional density estimates do not necessarily imply an unbiased SDF estimate, a shortcoming that is usually ignored in the literature.

We show that the SDF is strictly monotonic if and only if (conditional) expected returns of a class of option trading strategies are increasing in the strike price. The class is uniquely identified by the property that the logarithm of the payout function is concave over the set of ending security prices (in the price range in which the payouts are positive). In addition to calls and puts, this class includes butterfly spreads, bullish call and bearish put spreads, and binary call and put options.<sup>6</sup> An increase in the strike price for any strategy in this class shifts, in the first-order stochastic dominance sense, the probability weighted payoffs to lower values of the SDF, resulting in lower prices and higher expected returns. The butterfly spread, for example, has a payoff only within the stock-price window defined by the lower and upper strikes, and its expected return is the inverse of a weighted average of the SDF within this window. As the strike is shifted, so is the window, and the resulting

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<sup>5</sup>Other papers, such as Constantinides (1992), Chernov (2003), Bakshi and Chen (2001), Rosenberg and Engle (2002) and Brennan, Liu, and Xia (2006), have directly estimated the SDF. The latter two do not rely on unobserved state variables, and are therefore best adapted to estimating a projected SDF.

<sup>6</sup>See Example 3 below for a definition of butterfly payouts. A bullish (bearish) call spread is long a call at one strike price and short a call at a higher (lower) strike price, with both options on the same stock and same expiration. A put spread is constructed similarly.

change in the expected returns reveals information about the SDF slope.<sup>7</sup> Our test of SDF monotonicity therefore reduces to the straightforward problem of testing whether expected returns of a class of trading strategies are increasing in strike price.

Our empirical approach is most closely related to the work in Coval and Shumway (2001) and Bakshi, Madan, and Panayotov (2010), both of which use option-strategy returns to test monotonicity properties. Coval and Shumway (2001) show that truncated correlations between returns and the SDF are negative if and only if conditional expected call and put option returns are increasing in strike price.<sup>8</sup> We show that there can be a large gap between the class of SDFs whose truncated correlations are all negative and the class whose SDFs are monotonically decreasing. We characterize the (larger) set of trading strategies such that monotonicity of expected returns in strike is equivalent to a downward sloping SDF. Bakshi, Madan, and Panayotov (2010) use returns on a set of strategies with “upside payoffs,” call options, binary cash calls, and kernel calls to identify SDFs that are increasing and convex in the right tail (for returns above some critical level).<sup>9</sup> This form of nonmonotonicity implies that expected returns are decreasing in strike for strikes above a critical level. Unlike our paper, their tests apply only to a particular form of nonmonotonicity (U-shaped in the right tail), whereas our tests apply to any form of nonmonotonicity, whether in the tails or in the interior of the graph. Furthermore, we characterize the entire set of strategies whose expected returns are increasing in strike, which allows us to give necessary and sufficient conditions for overall monotonicity in terms of expected returns of option strategies. A key advantage of our approach, as with Coval and Shumway (2001) and Bakshi, Madan, and Panayotov (2010), is that it is model-free: We make no assumptions about the objective or

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<sup>7</sup>Though the case of butterfly spreads (as well as call options) suggests a link to skewness, we emphasize that a (first-order dominance) shift to the right in payouts does not imply, nor is implied by, an increase in skewness. For example, vanilla and binary put options both have log-concave payouts and, therefore, expected returns increasing in strike, but the skewness of returns of both typically decrease as the strike increases.

<sup>8</sup>That is, correlations conditional on the return exceeding any fixed amount, or falling below any fixed amount.

<sup>9</sup>Kernel calls are synthesized derivatives with payoff equal to the value of the projected kernel, or SDF, if the return exceeds some strike  $K$ , and zero otherwise.

risk-neutral distributions, and we have no need to estimate the SDF itself. Unlike Coval and Shumway (2001) and Bakshi, Madan, and Panayotov (2010), our approach can be used to statistically test overall SDF monotonicity.

The only other study we are aware of that examines SDF monotonicity of individual stocks is Ni (2007), who applies Coval and Shumway (2001) to test for weak SDF monotonicity by testing monotonicity of call expected returns with respect to strike price. Inconsistent with weak monotonicity, she finds that deep out-of-the-money (OTM) calls on average earn significantly lower average returns than deep in-the-money (ITM) calls. She argues that the evidence is consistent with risk-seeking behavior; in particular, a preference for idiosyncratic skewness induces a premium for deep OTM options. Our results suggest that her results are caused by a violation of *ceteris paribus*: the returns across consecutive strike groups in her study do not always correspond to the same underlying stock, with the problem most severe in the OTM strike groups.<sup>10</sup> Once *ceteris paribus* is upheld, by always differencing returns across strike while holding fixed the underlying stock, the anomaly disappears, and the average individual stock SDF is found to satisfy monotonicity.

We also apply our monotonicity tests to the projected S&P 500 SDF. Consistent with our model and the empirical fact that the bulk of stocks exhibit a downward-sloping implied smile (consistent with potential negative jumps), we find evidence of nonmonotonicity corresponding to negative returns. Average call returns are decreasing in strike for low (ITM) strikes (statistically significant at the 5% level for the second strike difference), and butterfly returns are decreasing in strike for the low strikes.<sup>11</sup> Both are consistent with a hump on the left side of the SDF graph. Our results, which are consistent with an SDF-return graph

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<sup>10</sup>As one indication, using her sorting rule the average elasticity for her deep OTM calls is smaller than that for the ATM group, even though Black-Scholes elasticities are monotonically increasing in strike, *ceteris paribus*. One explanation for this is that her sorting rule results in assigning a disproportionate number of low priced, and consequently, significantly higher volatility stocks to higher strike groups. In the Black-Scholes model, for example, the elasticity of a call option, and therefore the call option expected return, decreases with volatility. This causes her deep OTM calls to have lower average returns than ATM calls. Detailed tables illustrating these results can be provided upon request.

<sup>11</sup>Our call results for the S&P 500 are different from those reported in Coval and Shumway (2001), who (based on the sample period 1990-1995) finds positive, though statistically insignificant, average call returns increments in strike.

that is not monotonic, are consistent with results of Ait-Sahalia and Lo (2000); Jackwerth (2000); Rosenberg and Engle (2002); Brown and Jackwerth (2004) and Shive and Shumway (2009).<sup>12</sup>

Our work is also related to a number of stochastic dominance results. SDF monotonicity is a key assumption underlying Dybvig (1988)'s notion of "cost efficient" portfolios, as well as stochastic dominance bounds on option prices. In a complete-markets setting, the results of Bernard and Boyle (2010) and Beare (2011) show that nonmonotonicity of the projected SDF implies the existence of a derivatives trading strategy that yields returns that first-order stochastically dominate the stock returns. As with all stochastic dominance results, additional distributional conditions and utility assumptions are needed to conclude that nonmonotonicity implies equilibrium mispricing. SDF monotonicity is also the key assumption underlying the incomplete-markets stochastic dominance option pricing bounds in Perrakis and Ryan (1984); Ritchken (1985); Levy (1985); Constantinides and Perrakis (2002 and 2006); and Constantinides, Jackwerth, and Perrakis (2009).<sup>13</sup> The pattern of bound violations may suggest a departure of monotonicity of a particular form (for example, systematic overpricing of out-of-the money calls would suggest a violation of monotonicity for high terminal prices), but these violations do not lend themselves to a statistical test of overall monotonicity. Moreover, unlike our paper, their results rely heavily on the precision of their estimate of the conditional density of stock returns.

The remainder of the paper is organized as follows. Section I develops the framework for empirical testing of SDF monotonicity using returns of option trading strategies. Section II presents the jump diffusion model and examines the implications for SDF monotonicity and the slope of implied volatility smile. Section III describes the dataset and Section IV presents the empirical results. Section V concludes the paper. Appendices A-D contain,

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<sup>12</sup>There is some contradictory evidence. For example, Barone-Adesi and Dall'O (2010)'s findings support strict monotonicity.

<sup>13</sup>In the latter three papers SDF monotonicity follows from assumed limits on trading positions that ensure that marginal utility is non-decreasing in the stock price. Ritchken (1985) and Levy (1985) also derive bounds assuming only non-negativity of the SDF.

respectively, proofs for Section I, proofs for Section II , derivations for the skewness adjusted t-statistics, and the details of the joint tests.

## I. SDF Monotonicity and Expected Option Returns

This section provides the framework for empirical testing of SDF monotonicity using returns of option trading strategies. The main result, in Proposition 4 below, shows that “strict monotonicity” (a strictly decreasing projected SDF as a function of return) is equivalent to expected returns increasing in strike for a class of option trading strategies. This class is characterized in Proposition 3 as payout functions which are log concave, and include, in addition to long call and put options, long butterfly spreads and binary cash-or-nothing calls and puts. Proposition 4 complements the test of Coval and Shumway (2001), who show that a weaker form of monotonicity (negative truncated correlations between the SDF and underlying returns) can be tested using call and put option returns (see Proposition 2 below). We want to stress that this weak form of monotonicity is not the same as strict monotonicity. For example, an SDF that is monotonically decreasing in the stock return in the right and left tails, but exhibits a hump for intermediate returns, may satisfy weak monotonicity (see Example 1 below for a simple case). But butterfly spread returns with strikes sufficiently close can potentially detect the hump because as the strikes of the butterfly are increased from the decreasing part of the SDF to the increasing part, the expected returns will first increase and then decrease.

Fixing the current time  $t$  and some payout date  $T$ , an SDF, or pricing kernel,  $m_{t,T}$  is a (time- $T$  measurable) random variable such that the current price  $p_t(X)$  of any time- $T$  payout  $X$  satisfies  $p_t(X) = E_t(Xm_{t,T})$ . If the number of states is finite, then the state- $s$  value of the SDF,  $m_{t,T}^s$ , can be interpreted as the per-unit-probability price of a claim to one unit of state- $s$  consumption. Technicalities aside, we know that a strictly positive SDF exists if and only if there is no arbitrage.<sup>14</sup> For example, if there exists an agent

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<sup>14</sup>In the finite-state setting, the result follows directly from the separating hyperplane theorem. Note that

with utility from current consumption,  $c_t$ , and time- $T$  wealth,  $W_T$ , given by  $U_t(c_t, W_T) = u(t, c_t) + E_t u(T, W_T)$ , then the first-order condition for optimality implies that there exists an SDF equal to the agent's marginal rate of substitution,  $m_{t,T} = u_c(T, W_T) / u_c(t, c_t)$ , and risk aversion is equivalent to  $m_{t,T}$  decreasing in  $W_T$ . If each agent  $i$  has a utility function of this form (with possibly different functions  $u$ ), standard aggregation arguments imply monotonicity of  $m_{t,T}$  in aggregate time- $T$  wealth.<sup>15</sup>

Define the (gross) expected return function  $R_t$  by  $R_t(X) = E_t(X) / E_t(m_{t,T}X)$  for any time- $T$  payout  $X$ ; therefore  $R_t(1)$  is the riskless return. Fix throughout some stock with time- $T$  price (with dividends reinvested)  $S_T$ . The conditional distribution function of  $S_T$  is denoted  $F_t(\cdot)$ , and is assumed throughout (for simplicity) to satisfy the following assumption:

**Standing Assumption** The conditional distribution function  $F_t$  is absolutely continuous, with strictly positive density on  $(0, \infty)$ .

We use the common notation  $x^+ = \max(0, x)$  for any  $x \in \mathbb{R}$ . Finally, we define the risk-neutral measure  $\tilde{P}$  at time 0 as usual via

$$\frac{d\tilde{P}}{dP} = \frac{m_{0,T}}{E(m_{0,T})}$$

(that is, the risk-neutral probabilities are obtained by scaling the SDF-weighted true probabilities), and denote by  $\tilde{E}$  the expectation operator under  $\tilde{P}$ .

For comparison with our later results, the following proposition gives some well-known equivalent characterizations of a positive risk premium.

**Proposition 1** *The following are equivalent:*

- i)  $R_t(S_T) > R_t(1)$ .
- ii)  $Cov_t(S_T, m_{t,T}) < 0$ .

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throughout we are considering the case of unconstrained trading (no short-selling constraints, for example).

<sup>15</sup>A sufficient condition is Pareto optimality of the equilibrium allocation. If the value of the S&P 500 represented aggregate wealth, the SDF projected on S&P 500 returns would be downward sloping in this setting.

$$iii) E_t(S_T) > \tilde{E}_t(S_T).$$

**Proof.** See the appendix. ■

The intuition for the equivalence of (i) and (ii) is that if higher stock payouts are associated with less valuable payout states, in the sense that a linear regression of  $S_T$  against  $m_{t,T}$  has a negative slope, then the current price is more heavily discounted, resulting in a higher expected return. The equivalence to (iii) follows because replacing the true probability for each state by the forward price per unit payout in that state reduces the expected payout if payouts are higher in low-value states.

A stronger condition than negative covariance, or a negative regression slope, is what we term weak monotonicity with respect to  $S_T$ , which we define by negative slopes for all regressions truncated to a half-line.

**Definition 1** *The SDF  $m_{t,T}$  is weakly monotonic with respect to  $S_T$  if*

$$Cov_t(S_T, m_{t,T} \mid S_T \in [\alpha, \beta]) < 0 \quad \text{for all } 0 \leq \alpha < \beta \text{ with either } \alpha = 0 \text{ or } \beta = \infty.$$

The following proposition relates weak monotonicity to monotonicity of option expected returns, and analogous to the unconditional case, characterizes weak monotonicity in terms of conditional risk premia and conditional risk-neutral expectations:

**Proposition 2 (weak monotonicity)** *The following are equivalent:*

- i)  $m_{t,T}$  is weakly monotonic with respect to  $S_T$ .*
- ii)  $R_t(S_T 1_{\{S_T \in [\alpha, \beta]\}}) > R_t(1_{\{S_T \in [\alpha, \beta]\}})$  for all  $0 \leq \alpha < \beta$  with either  $\alpha = 0$  or  $\beta = \infty$ .*
- iii)  $E_t(S_T \mid S_T \in [\alpha, \beta]) > \tilde{E}_t(S_T \mid S_T \in [\alpha, \beta])$  for all  $0 \leq \alpha < \beta$  with either  $\alpha = 0$  or  $\beta = \infty$ .*
- iv) (Coval and Shumway (2001))  $R_t((S_T - K)^+)$  and  $R_t((K - S_T)^+)$  are strictly increasing in  $K$ .*

**Proof.** See the appendix. ■

Part (ii) says that weak monotonicity is equivalent to strict positivity of conditional risk premia; that is, positive risk premia for all asset-or-nothing binary calls (puts) relative to cash-or-nothing binary calls (puts). Part (iv) is the Coval and Shumway (2001) characterization in terms of traded option strategies, which they apply to testing weak monotonicity of  $m_{t,T}$  with respect to the S&P 500 index.

Finally, we define strict monotonicity as monotonicity of the conditional expectation, or nonlinear regression of  $m_{t,T}$  on  $S_T$ :

**Definition 2** *The SDF  $m_{t,T}$  is strictly monotonic with respect to  $S_T$  if*

$$m_{t,T}(s) = E_t(m_{t,T} | S_T = s) \text{ is strictly decreasing in } s.$$

While it might appear that weak monotonicity is close to strict monotonicity, the following simple example shows that weak monotonicity can hold despite the absence of strict monotonicity for intermediate stock prices.<sup>16</sup>

**Example 1** *Suppose a discrete setting with four equally-likely and equally-spaced possible terminal stock prices:  $s_1 < s_2 < s_3 < s_4$ . Then  $m_{t,T}$  is weakly monotonic if and only if  $m_{t,T}(s_1) > m_{t,T}(s_4)$ ; and  $m_{t,T}(s_2), m_{t,T}(s_3) \in (m_{t,T}(s_1), m_{t,T}(s_4))$ . That is, the price of consumption in the lowest stock-price state exceeds that in the highest stock-price state, and the prices of consumption in the two middle states are somewhere in between. Therefore weak monotonicity can hold even though  $m_{t,T}(s_2) < m_{t,T}(s_3)$ ; furthermore,  $m_{t,T}(s_3)$  can exceed  $m_{t,T}(s_2)$  by almost  $m_{t,T}(s_1) - m_{t,T}(s_4)$ .*

Strict monotonicity obviously implies weak monotonicity, and therefore a violation of monotonicity of conditional expected call or put returns with respect to the strike price

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<sup>16</sup>A derivation is provided in the appendix. We relax the standing assumption for the purpose of the example.

would contradict strict monotonicity of  $m_{t,T}$ . However, a larger family of trading strategies must be considered to affirm strict monotonicity. We obtain such a family of trading strategies in terms of a class of payoff functions  $g(\cdot)$  which satisfy some (mostly technical) regularity conditions.

**Definition 3** *A payoff function  $g$  is regular if  $g(x) > 0$  for  $x \in (x_1, x_2)$ , where  $-\infty \leq x_1 < x_2 \leq \infty$ , and  $g(x) = 0$  for  $x \notin [x_1, x_2]$ .<sup>17</sup> Furthermore,  $g(x)$  is continuous for  $x \in (x_1, x_2)$  and  $g'(x)$  is piecewise continuous.*

The following proposition characterizes the class of regular payoff functions  $g$  such that the conditional expected return  $R_t(g(S_T - K))$  is increasing in the shift parameter (or, strike price)  $K$  for all stock-price distribution functions and any strictly monotonic SDF.

**Proposition 3** *Suppose  $m_{t,T}$  is strictly monotonic and the payoff function  $g$  is regular. Then the conditional expected return  $R_t(g(S_T - K))$  is increasing in  $K$  for all conditional distribution functions  $F_t$  if and only if*

$$\ln(g(x)) \text{ is concave in } x \text{ for all } x \in (x_1, x_2), \quad (1)$$

where  $(x_1, x_2)$  is the support of  $g$ .<sup>18</sup>

**Proof.** See the appendix. ■

For any strictly decreasing  $m_{t,T}(\cdot)$ , the only class of regular payoffs with conditional expected returns monotonic in strike (for any conditional distribution function) is that characterized by log concavity; that is,  $g'(x)/g(x)$  is strictly decreasing in  $x$  over the support of  $g$ .<sup>19</sup> If a payoff is regular and piecewise linear (as in the case of portfolios of stock and

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<sup>17</sup>That is, the support of  $g$  is  $(x_1, x_2)$ .

<sup>18</sup> $R_t(g(S_T - K))$  is *strictly* increasing in  $K$  under the condition of the proposition unless the payoff function is within the exponential class:  $g(x) = \exp(\alpha + \beta x)$ , for some  $\alpha, \beta \in \mathbb{R}$ . Within this class  $R_t(g(S_T - K))$  is invariant to  $K$ .

<sup>19</sup>More precisely, the left and right-hand derivatives of  $\ln(g(\cdot))$  (which exist by concavity) are strictly decreasing. Concave functions are differentiable (the left and right-hand derivatives are the same) almost everywhere.

traded options), log concavity is equivalent to concavity of  $g$  within the support of  $g$ . All vanilla options and all binary options satisfy log concavity because their slopes are constant in their positive-payoff region. Butterfly spreads, bullish call, and bearish put spreads also satisfy log concavity. On the other hand, long straddles violate log concavity because the slope increases from  $-1$  to  $1$  as the stock price moves from below to above the strike (short straddles violate regularity). As shown in the proof, log concavity of the payoff implies that an increase in the strike shifts (in the first-order dominance sense) the probability-weighted payoffs to higher stock prices; the lower values of the SDF corresponding to higher stock prices implies a lower price (per unit probability-weighted payoff) and therefore a higher conditional expected return.

Proposition 4 characterizes strict monotonicity in the analogous way that Proposition 2 characterized weak monotonicity. The main result is part (v) which shows that monotonicity of conditional expected returns in strike for this payoff class is equivalent to strict monotonicity of  $m_{t,T}$ .

**Proposition 4 (strict monotonicity)** *Suppose  $m_{t,T}()$  is differentiable. Then the following are equivalent:*

- i)  $m_{t,T}$  is strictly monotonic with respect to  $S_T$ .*
- ii)  $\text{Cov}_t(S_T, m_{t,T} \mid S_T \in [\alpha, \beta]) < 0$  for all  $0 \leq \alpha < \beta$ .*
- iii)  $R_t(S_T 1_{\{S_T \in [\alpha, \beta]\}}) > R_t(1_{\{S_T \in [\alpha, \beta]\}})$  for all  $0 \leq \alpha < \beta$ .*
- iv)  $E_t(S_T \mid S_T \in [\alpha, \beta]) > \tilde{E}_t(S_T \mid S_T \in [\alpha, \beta])$  for all  $0 \leq \alpha < \beta$ .*
- v)  $R_t(g(S_T - K))$  is strictly increasing in  $K$  for all regular payout functions  $g$  satisfying the log-concavity condition (1).*

**Proof.** See the appendix. ■

Part (ii) shows that strict monotonicity is equivalent to negative slopes for all regressions truncated on both sides; that is, all regressions conditional on the stock price falling within any interval. Part (iv) shows the equivalence to the positivity of all conditional risk premia,

with payouts confined to arbitrary intervals; or, equivalently, the positivity of risk premia of asset-or-nothing versus cash-or-nothing binary options with payouts contingent on the stock price falling within an arbitrary interval. Because binary options were not exchange traded throughout our sample period,<sup>20</sup> the characterization in (v) provides a better empirical framework for testing monotonicity because we can rely on traded option strategy returns.

The examples that follow satisfy the log-concavity condition (1). Strict monotonicity of  $m_{t,T}$  therefore implies monotonicity of conditional expected returns in the strike price..

**Example 2 (Call and put options)** *The call and put payoff functions are  $g(x) = x^+$  and  $g(x) = (-x)^+$ , respectively. The inverse of the expected return for the call (put) is a weighted average of the right (left) tail of the SDF:*

$$\frac{1}{R_t((S_T - K)^+)} = E_t \left( \frac{(S_T - K) 1_{\{S_T \geq K\}}}{E_t((S_T - K) 1_{\{S_T \geq K\}})} m_{t,T}(S_T) \right),$$

$$\frac{1}{R_t((S_T - K)^-)} = E_t \left( \frac{(K - S_T) 1_{\{S_T \leq K\}}}{E_t((K - S_T) 1_{\{S_T \leq K\}})} m_{t,T}(S_T) \right).$$

As the strike price increases, the weight function shifts to the right covering only a shrinking tail region of  $m_{t,T}()$  to the right in the case of calls, and an expanding tail region to the left in the case of puts. When  $m_{t,T}()$  is decreasing, the weighted average decreases in either case, increasing the expected return.

The next example considers symmetric butterfly spreads. Note that asymmetric butterfly spreads also satisfy log-concavity.

**Example 3 (Butterfly spread)** *A strategy of long one call each at strike prices  $K - \Delta K$  and  $K + \Delta K$ , and short two calls at strike  $K$ , has the payoff function  $g(x) = (\Delta K - |x|)^+$ . The inverse of the conditional expected return represents a weighted average of  $m_{t,T}(s)$  with*

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<sup>20</sup>The CBOE introduced binary options on SPX and VIX indexes in July 2008.

strictly positive weights only for  $s \in (K - \Delta K, K + \Delta K)$ :

$$\frac{1}{R_t((\Delta K - |S_T - K|)^+)} = E_t \left( m_{t,T}(S_T) \frac{(\Delta K - |S_T - K|)^+}{E_t\{(\Delta K - |S_T - K|)^+\}} \right).$$

The weights with the butterfly spread are concentrated in the window between the lower and upper strike, causing the weighted average to decline, if  $m_{t,T}()$  is decreasing, as the window is shifted to the right. Note that in the limit, as the strike increments goes to zero, the inverse of the expected return equals the value of the SDF at  $K$ : that is,  $\lim_{\Delta K \rightarrow 0} \frac{1}{R_t((\Delta K - |S_T - K|)^+)} = m_{t,T}(K)$ .

## II. Jump-Diffusion Models and SDF Monotonicity

In this section we examine SDF monotonicity in the context of a simple jump-diffusion model with a common source of “crash” risk, and both systematic and idiosyncratic components of Brownian risk. We show that whether we are considering an individual stock or a stock index, if the equilibrium per-unit-return compensation for Brownian risk exceeds that of jump risk, a hump in the SDF can result. If the Brownian compensation is smaller, on the other hand, the downward slope in the SDF is even steeper than the Black-Scholes SDF. SDF humps are more likely when (1) the jump component of return is large relative to the volatility of the Brownian component of return, and (2) the market price of Brownian risk is high relative to the market price of jump risk. Because diversification amplifies both factors, thus increasing the per-unit-return Brownian compensation relative to the jump compensation, SDF humps are more likely for the index than the typical stock. The relative jump component of return can be measured using the implied volatility smile. Using the approach of Yan (2011), we show that the (normalized) slope of the ATM implied volatility smile near expiration equals the ratio of the risk-neutral expected jump component of return to the volatility of the Brownian component of return (consistent with the special case derived in Yan (2011)). For example, if the downside jump component of return is large

relative to the Brownian component, then our model predicts a steeper downward sloping smile and more likely violations of SDF monotonicity corresponding to negative returns. We use this property in Section IV to sort stocks according to downside and upside jump potential; we then investigate whether, consistent with the theory in this section, return jumps explain SDF monotonicity violations. Note however that the smile-steepness and SDF-nonmonotonicity phenomena are not the same. The implied smile slope depends only on the risk-neutral expected jump return relative to Brownian diffusion, whereas SDF violations depend critically on the market prices of risk, as well as the concentration of jump risk.

The main theoretical result of this section is Proposition 5, which, for a general jump-diffusion model, characterizes monotonicity of the SDF for both individual stocks and a fixed-weight stock index. The general jump diffusion model also highlights the role of common versus idiosyncratic jump components of return. Local index SDF violations are more likely with common jump components. The intuition for most of the general results is captured by a simple jump diffusion model, which is similar to that used by Branger and Schlag (2004) to explain the steepness of the index smile.<sup>21</sup> The main results for the simple model appear in Lemma 1, which we present first and is mostly a corollary to Proposition 5.

We fix throughout the market price of Brownian risk  $\eta$ , such that  $d\tilde{B}_t^c = dB_t^c + \eta^c dt$  is standard Brownian motion under the risk-neutral measure  $\tilde{P}$ , and a market price per unit jump risk, which in the fixed-jump model is equal to  $\tilde{\lambda} - \lambda$ , the difference in the risk-neutral and physical jump intensities. That is, we hold fixed the (unprojected) state-price density, and examine the implications of the stock-return models on the projected individual-stock and stock-index SDFs.<sup>22</sup> We assume  $n > 1$  stocks, with  $S^i$  denoting the price process of stock  $i$ . The stock index  $S^\phi$  is constructed using the fixed weights  $(\phi^1, \dots, \phi^n)$  satisfying  $\sum_{i=1}^n \phi^i = 1$ . We choose a fixed-weight index for tractability, but the properties of this portfolio are very similar to those of an additive index (with fixed number of shares over

<sup>21</sup>Index versus individual-stock smile steepness is evaluated using numerical simulations in their paper.

<sup>22</sup>The existence of a strictly positive state-price density implies that the model is arbitrage free.

time) for the return horizons we consider.<sup>23</sup>

It is convenient in this section to express each SDF in terms of asset  $i$ 's log return from  $t$  to  $T$ , which we denote  $x = \ln(S_T^i/S_t^i)$ . Henceforth we fix  $t$  and  $S_t^i$  and define the SDF of asset  $i$ 's log return as

$$m^i(x) = m_{t,T}^i(S_t^i e^x) \quad i = \phi, 1, \dots, n;$$

that is,  $m^i(\ln(S_T^i/S_t^i)) = m_{t,T}^i(S_t^i)$ . We use  $\tau = T - t$  to denote the length of the time interval over which the SDF is computed. Finally, we denote by  $\hat{\sigma}^i(K)$  the Black-Scholes implied volatility for asset  $i$  at strike price  $K$ .

## A. A Simple Model

Similar to Branger and Schlag (2004), suppose that returns of the  $i$ th stock satisfy

$$\frac{dS_t^i}{S_{t-}^i} = \mu^i dt + \sigma^i \left( \rho dB_t^c + \sqrt{1 - \rho^2} dB_t^i \right) + k^i dN(t), \quad i = 1, \dots, n, \quad (2)$$

where<sup>24</sup>  $\rho \in [0, 1)$ ,  $\sigma^i > 0$  all  $i$ ,  $\{B^c, B^1, \dots, B^n\}$  are mutually independent standard Brownian processes, and  $N(t)$  is a Poisson process (independent of the Brownian processes) with intensities  $\lambda$  and  $\tilde{\lambda}$  under the measures  $P$  and  $\tilde{P}$ . The common source of jumps is  $N$ , though jumps in  $N$  can have different impacts on different stocks (the  $k^i$ 's need not all be negative). When interpreting the results, we typically assume a positive price of jump risk,  $\tilde{\lambda} > \lambda$ , a positive price of common diffusion risk,  $\eta^c > 0$ , and downward price movements at jumps,  $k^i < 0$ . The idiosyncratic Brownian processes are unpriced ( $\eta^i = 0$ ):  $\{B^1, \dots, B^n\}$  are standard Brownian processes under  $\tilde{P}$  as well as  $P$ .

The expression for the stock-index return is simplified by introducing a new standard Brownian process  $B^\phi$ , independent of  $B^c$  and  $N$ , to represent the idiosyncratic part of the

<sup>23</sup>We confirm this through simulations for a one month investment horizon and we find that the return distribution for a fixed-weight index and a fixed share index are virtually indistinguishable

<sup>24</sup>The assumption  $\rho \geq 0$  is without loss of generality (we can always reverse the signs  $B^c$ ); we assume  $\sigma^i > 0$  for simplicity, so that each stock has positive per-unit-return market price of systematic Brownian risk.

index return.<sup>25</sup> The stock-return dynamics (2) imply that the stock-index returns satisfy

$$\frac{dS_t^\phi}{S_{t-}^\phi} = \mu^\phi dt + \sigma^\phi \left( \rho^\phi dB_t^c + \sqrt{1 - (\rho^\phi)^2} dB_t^\phi \right) + k^\phi dN(t) \quad (3)$$

where

$$\begin{aligned} \mu^\phi &= \sum_{i=1}^n \phi^i \mu^i, \quad k^\phi = \sum_{i=1}^n \phi^i k^i, \\ \sigma^\phi &= \sqrt{\rho^2 \left( \sum_{i=1}^n \phi^i \sigma^i \right)^2 + (1 - \rho^2) \sum_{i=1}^n (\phi^i \sigma^i)^2}, \quad \rho^\phi = \frac{\rho}{\sigma^\phi} \sum_{i=1}^n \phi^i \sigma^i. \end{aligned} \quad (4)$$

The parameter  $\rho^\phi$  represents the correlation of the index returns with the common risk component, and satisfies  $\rho^\phi > \rho$ . In the limiting case of a “well-diversified” portfolio, the idiosyncratic part of the index return vanishes and  $\rho^\phi \rightarrow 1$ .<sup>26</sup>

The following lemma gives the SDF expression and a necessary and sufficient condition for SDF monotonicity. It also gives an expression for the ATM implied slope for small  $\tau$ . All the expressions apply to both individual stocks ( $i \in \{1, \dots, n\}$ ) as well as the index ( $i = \phi$ ). Note that the SDF in the B-S model is<sup>27</sup>  $m_{BS}^i(x) \propto \exp(-x\rho^i\eta^c/\sigma^i)$ . Finally, define  $\delta^i$  as the difference in the per-unit-return compensations for Brownian and jump risks, respectively:

$$\delta^i = \frac{\rho^i \eta^c}{\sigma^i} - \frac{\ln(\tilde{\lambda}/\lambda)}{-\psi^i}; \quad (5)$$

**Lemma 1** *Let  $\rho^i = \rho$  for each individual stock  $i = 1, \dots, n$ ; and for each  $i \in \{\phi, 1, \dots, n\}$  define the log jump return  $\psi^i = \ln(1 + k^i)$ .*

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$$^{25} B^\phi \text{ satisfies } dB_t^\phi = \left( \sum_{i=1}^n (\phi^i \sigma^i)^2 \right)^{-1/2} \sum_{i=1}^n \phi^i \sigma^i dB_t^i.$$

<sup>26</sup>A well-diversified portfolio is the limit of a sequence of  $n$ -stock portfolios as  $n \rightarrow \infty$ , assuming the maximum absolute investment weight in the sequence converges to zero, and also assuming a uniform upper and a strictly positive lower bound on the  $\sigma^i$ 's.

<sup>27</sup>See the proof in the appendix for the proportionality constant, which doesn't affect monotonicity.

a) For any  $x \in \mathbb{R}$ , the SDF at  $\ln(S_T/S_t) = x$  is given by

$$m^i(x) \propto m_{BS}^i(x) E \left( e^{\delta^i \psi^i N_\tau} \middle| \ln(S_\tau/S_0) = x \right). \quad (6)$$

It follows that the log SDF slope satisfies

$$\delta^i \gtrless 0 \implies \frac{d}{dx} \ln m^i(x) \gtrless \frac{d}{dx} \ln m_{BS}^i(x), \quad \text{all } x \in \mathbb{R}, \quad i \in \{\phi, 1, \dots, n\}. \quad (7)$$

Furthermore,  $\delta^i > 0$  implies that for sufficiently small  $\tau$ , asset  $i$ 's SDF is nonmonotonic in a neighborhood of  $\psi^i/2$ .

b) Monotonicity at  $x$  is characterized by

$$\frac{d}{dx} m^i(x) < 0 \iff \frac{\psi^i}{\sigma^i} \left\{ \tilde{E}_t(N(T) - N(t) | x) - E_t(N(T) - N(t) | x) \right\} < \rho^i \eta^c \tau. \quad (8)$$

c) The sensitivity of the slope of the log SDF with respect to the market price of risk is

$$\frac{d}{d(\rho^i \eta^c)} \frac{d}{dx} \ln \{m^i(x)\} = -\frac{1}{\sigma^i} \left( 1 - \frac{1}{\tau} \left( \frac{\psi^i}{\sigma^i} \right)^2 \text{Var}_t \{N(T) - N(t) | x\} \right). \quad (9)$$

d) The normalized slope of the implied volatility curve at the strike price  $K = e^{r\tau} S_t^i$  satisfies<sup>28</sup>

$$K \frac{d}{dK} \hat{\sigma}^i(K) = \frac{\tilde{\lambda} k^i}{\sigma^i} + O(\sqrt{\tau}). \quad (10)$$

**Proof.** See the appendix. ■

The following heuristic argument explains the impact of jumps on the SDF. Fix some stock or index  $i$ . Suppose we are very close to expiration, and want to assess the value of the SDF at some log return  $y$ . Assume for simplicity that  $\rho^i = 1$ ,  $\psi^i < 0$ , and  $\tilde{\lambda} > \lambda$  (a positive price to “crash” risk), and we restrict  $N$  to at most one jump before expiration.<sup>29</sup> Recall

<sup>28</sup>That is, there exist constants  $\gamma, \varepsilon > 0$  such that  $\left| K \frac{d}{dK} \hat{\sigma}^i(K) - \frac{\tilde{\lambda} k^i}{\sigma^i} \right| \leq \gamma \sqrt{\tau}$  for all  $\tau < \varepsilon$ .

<sup>29</sup>Allowing more than one jump results in SDF nonmonotonicities (for sufficiently small  $\tau$ ) for  $y$  close to  $\psi^i/2$  for any whole number  $k$ , though the violations for  $k > 1$  are likely to be too far away from the money

that the market price of the systematic Brownian risk is  $\eta^c$ , and it is well known that the instantaneous price of downward jump risk is  $\ln(\tilde{\lambda}/\lambda)$ .<sup>30</sup> For any return  $y > \psi^i/2$  over the next instant, the conditional probability of a jump component to that return is near zero, and therefore the log SDF is  $\ln m(y) \approx -y\eta^c/\sigma^i$  (because the Brownian increment must be  $\Delta B^c = y/\sigma^i$ ), as in the B-S model. On the other hand, for any return  $y < \psi^i/2$ , the conditional probability of a jump is near one, and there are two components to the return compensation:

$$\ln m(y) \approx -(y - \psi^i)\eta^c/\sigma^i + \ln(\tilde{\lambda}/\lambda), \quad y < \psi^i/2.$$

The first part is the compensation for the Brownian increment  $\Delta B^c = (y - \psi^i)/\sigma^i$ , and the second for the jump  $\Delta N_t$ . If  $\delta^i = 0$  (equal compensations per unit return) then the SDF is unaffected by the jump risk, and  $\ln m(y) \approx -y\eta^c/\sigma^i$  for *all*  $y$ . If  $\delta^i > 0$  then as  $y$  increases from  $\psi^i/2 - \varepsilon$  to  $\psi^i/2 + \varepsilon$  for small  $\varepsilon > 0$ , the SDF increases by  $\delta^i$  (resulting in a hump), because the return attributable to both jump and diffusion risk switches to return attributable solely to (higher priced) diffusion risk.<sup>31</sup> Finally,  $\delta^i < 0$  implies a steeper downward slope to the SDF than B-S, particularly near  $y = \psi^i/2$ .

As the time to expiration increases, the story becomes more complicated, and the SDF becomes more smooth around  $y = \psi^i/2$ . But the intuition remains intact: monotonicity violations are caused by disparate pricing of jump risk and diffusion risk, with violations occurring near returns where there is high uncertainty over which source of risk caused the return. Part (a) shows that  $\delta^i > 0$  is a necessary and sufficient condition for an SDF hump existing for some maturity  $\tau > 0$ . If  $\delta^i = 0$  then the SDF is proportional to the B-S SDF for every  $\tau > 0$ , and  $\delta^i < 0$  implies a more steeply downward-sloping log SDF than B-S. Parts (b) and (c) show that violations are more likely the shorter the time to expiration, and

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to be detected with option-pricing data. Because the SDF represents the return compensation *per unit probability*, unlikely outcomes (the probability of more than one jump over small  $\tau$  is of order  $\tau^2$ ) cannot always be ignored.

<sup>30</sup>The term  $-\ln(\lambda/\tilde{\lambda})$  is the coefficient of  $dN_t$  in the Ito expansion of the state-price density, and  $-\eta^c$  which is the coefficient of  $dB_t^c$ .

<sup>31</sup>If  $\tilde{\lambda} > \lambda$  but stock  $i$ 's price *increases* at market crashes ( $\psi^i > 0$ ), then  $\delta > 0$  necessarily holds because the positive crash reward (instead of a negative Brownian reward) is associated with returns above  $\psi^i/2$ .

the larger the per-unit-return compensation for Brownian versus jump risk. In the typical case of  $\tilde{\lambda} > \lambda$  and  $\psi^i < 0$ , then a hump on the left side of the graph is possible in the vicinity of  $\psi^i/2$  if  $\delta^i > 0$ ; but for returns to the right of  $\psi^i/2$ , the SDF will increasingly resemble the B-S SDF.<sup>32</sup> The case of  $\tilde{\lambda} < \lambda$  and  $\psi^i > 0$  (priced positive jumps) is symmetric:  $\delta^i > 0$  implies a possible hump on the right side of the SDF graph, and an SDF resembling the B-S SDF on the left side of the graph. Parts (b) and (c) relate SDF humps to conditional uncertainty about the number of jumps.<sup>33</sup> The expression (9) shows that increasing  $\rho^i \eta^c$  can result in an *increase* in the SDF slope if  $|\psi/\sigma^i|$  and the conditional variance of the number of jumps are sufficiently large. For example, if  $\psi = -10\%$  and  $\tau$  is small, then the variance of  $N(T) - N(t)$  conditional on  $y = -5\%$  will be close to its limiting (as  $\tau \rightarrow 0$ ) maximum  $1/4$  (because  $N(T) - N(t) \in \{0, 1\}$  will be close to equally likely). The source of the violation is also seen in (8), because the expected number of jumps conditional on  $y$  for  $y$  close to  $\psi^i/2$  can be sensitive to whether the expectation is under  $P$  or  $\tilde{P}$ .

Part (d) of the Lemma shows that the normalized ATM implied volatility slope near expiration equals the ratio of the expected risk-neutral jump return to diffusion coefficient, and can therefore measure the relative jump-to-diffusion risks for both stocks and the index. Diversification of Brownian risk results in a smaller denominator (smaller  $\sigma^i$ ) for the index than the average stock, resulting in a steeper normalized index smile. A large absolute jump-to-diffusion-risk ratio can be interpreted as essentially a necessary condition for an SDF violation, because part (a) of the lemma shows that an SDF hump is more likely for large  $\rho^i |\ln(1 + k^i)| / \sigma^i$ .

The key differences in this model between individual stocks and the index are 1) the log-jump to diffusion-volatility ratio  $|\psi^i| / \sigma^i$  tends to be larger for the stock index than the average individual stock, and 2) the compensation per unit Brownian exposure  $\rho^i \eta^c$  is larger

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<sup>32</sup>More precisely, as  $(y - \psi^i/2) / \sqrt{\tau}$  becomes large (small), the  $j = 0$  terms in the summations dominate and the SDF converges to the B-S SDF. On the other hand, for  $y$  close to  $\psi^i/2$  then the first two terms in summations dominate, which can result in nonmonotonicity.

<sup>33</sup>At first glance (8) seems to suggest that increasing  $\rho^i \eta^c$  makes monotonicity *more* likely. But as explained below increasing  $\rho^i \eta^c$  also increases physical stock drift relative the the risk-neutral drift., causing the physical and risk-neutral conditional expectations to diverge.

for the index than individual stocks (recall  $\rho^\phi > \rho$ ). Both factors imply more likely local monotonicity violations for the index than the individual stocks, even though the index SDF will tend to be steeper away from the hump.<sup>34</sup>

In a representative-agent setting, with aggregate wealth equal to the index value  $S^\phi$ , the per-unit-return Brownian and jump compensations for the index must match: the agent's optimality condition combined with market clearing implies  $\delta^\phi = 0$  because both terms on the right side of (5) equal to the agent's coefficient of relative risk aversion. The index SDF will therefore essentially match the B-S SDF, and the typical stock will be even more steeply downward sloping than the B-S SDF. (An individual stock with a large absolute jump could exhibit a hump, however.) Outside the representative-agent setting, however, equality of the Brownian and jump compensations would be merely coincidental.

Our simple model ignores diversification of jump risk: all return jumps are assumed to arrive simultaneously. The difference between the index SDF and the typical individual-stock SDF in this model is driven by the ability to diversify the idiosyncratic part of the Brownian risk, and the inability to diversify jump risk. It is simple to apply the above results to the model of Branger and Schlag (2004), in which jumps also have an idiosyncratic component. Suppose

$$\frac{dS_t^i}{S_{t-}^i} = \mu^i dt + \sigma^i \left( \rho dB_t^c + \sqrt{1 - \rho^2} dB_t^i \right) + k^i \left( dN^c(t) + dN^i(t) \right), \quad i = 1, \dots, n, \quad (11)$$

with all the counting processes independent, intensities  $\lambda^c$  and  $\lambda^i$ , and risk-neutral intensities  $\tilde{\lambda}^c$  and  $\tilde{\lambda}^i = \lambda^i$ . (Section B allows for more general specifications.) By reinterpreting  $N^c + N^i$  as a single jump with intensity  $\lambda^c + \lambda^i$ , then the results of Lemma 1 (a) apply to stocks and

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<sup>34</sup>On the other hand, the definition  $\psi^\phi = \ln(1 + k^\phi)$  together with Jensen's inequality implies:

$$\psi^\phi \geq \sum_{i=1}^n \phi^i \psi^i.$$

For example, if the individual-stock log jumps were all negative, the index log jump would be closer to zero than the weighted average.

a “well-diversified” index (with no idiosyncratic risk) after replacing (5) with

$$\delta^i = \frac{\rho\eta^c}{\sigma^i} - \frac{\ln\left(\frac{\tilde{\lambda}^c + \lambda^i}{\lambda^c + \lambda^i}\right)}{-\psi^i}, \quad i = 1, \dots, n; \quad \delta^\phi = \frac{\eta^c}{\sigma^\phi} - \frac{\ln\left(\frac{\tilde{\lambda}^c}{\lambda^c}\right)}{-\psi^\phi}.$$

An individual stock is more likely to exhibit a hump if the absolute jump size is large relative to  $\sigma^i$ , and jumps are mostly idiosyncratic ( $\lambda^i \gg \tilde{\lambda}^c$ ).

Whether the typical stock SDF behaves better or worse than the index SDF in the model (11) depends on the compensation for systematic jump risk and the extent to which the Brownian and jump risks are diversifiable. For example, with identical stock  $\sigma^i$ 's, and  $\psi^i$ 's (assume  $\psi^i < 0$ ), then each individual stock would be “better behaved,” in the sense that  $\delta^\phi \leq 0$  implies  $\delta^i \leq 0$ , if

$$\frac{1}{\rho} \ln\left(\frac{\tilde{\lambda}^c + \lambda^i}{\lambda^c + \lambda^i}\right) > \ln\left(\frac{\tilde{\lambda}^c}{\lambda^c}\right), \quad (12)$$

((12) is equivalent to  $\delta^\phi > \delta^i/\rho$ ) which is more likely if  $\rho$  is small (relatively large diversifiable component of Brownian risk) and  $\lambda^i$  is small (relatively large common component to jump risk). Under the same assumptions (but for any index), the ratio of stock  $i$ 's normalized implied slope to the normalized slope of the index is  $\rho\left(\tilde{\lambda} + \lambda^i\right) / \sum_{i=1}^n \phi^i\left(\tilde{\lambda} + \lambda^i\right)$  (recall  $\rho^i = \rho < 1$  for  $i = 1, \dots, n$ ), which is satisfied for the typical stock regardless of how what fraction of each stock's total intensity is systematic. The smile-slope result is caused by Brownian diversification, and is unaffected by jump diversification.

Figures 1 - 3 illustrate some of the results. Figure 1 illustrates the steepening hump in the SDF as the jump risk becomes more severe: the hump begins to appear with a jump return of about  $k = -25\%$ , centered in the vicinity of  $\psi/2 = -14.4\%$  (where conditional uncertainty of the jump contribution is highest). Figure 2 shows the impact of the market-price-of-Brownian-risk parameter  $\eta^c$ : monotonicity is satisfied (though with a shallow slope) for small  $\eta^c$ . As  $\eta^c$  increases, a hump arises (again in the vicinity of  $k/2$ ) even though the SDF slope becomes more negative in the positive-return region.<sup>35</sup> Figure 3 illustrates the

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<sup>35</sup>The parameters for the plots are chosen in a manner consistent with the observed data. For example

impact of diversification on SDF monotonicity. Plot A illustrates a single stock ( $N = 1$ ), which obviously satisfies monotonicity. We construct equally-weighted portfolios of stocks with the same parameters, and increase  $N$  (plots B-D). Even the two-stock portfolio ( $N = 2$ ) begins to exhibit a hump, which becomes more pronounced as  $N$  increases.<sup>36</sup>

## B. A More General Jump-Diffusion Model

We allow a general specification of the jump component of returns by assuming

$$S_t^i = S_0^i \exp \left\{ \left[ \mu^i - \frac{1}{2} (\sigma^i)^2 \right] t + \sigma^i \left( \rho B_t^c + \sqrt{1 - \rho^2} B_t^i \right) \right\} J_t^i, \quad i = 1, \dots, n, \quad (13)$$

where  $\{\ln(J_t^i); t \geq 0\}$  is assumed to be a Levy process (stationary independent increments) independent of  $B$  for all  $i$ . The model of Section A corresponds to  $J_t^i = (1 + k^i)^{N(t)}$  (and  $\ln \left\{ \tilde{E}(J_1^i) \right\} = \tilde{\lambda} k^i$ ).<sup>37</sup> Note that given any individual-stock jump processes  $(J^1, \dots, J^n)$ , the index jump process  $J^\phi$  satisfies<sup>38</sup>

$$\frac{dJ_t^\phi}{J_{t-}^\phi} = \sum_{i=1}^n \phi^i \frac{dJ_t^i}{J_{t-}^i}. \quad (14)$$

(which yields  $J_t^\phi = (1 + k^\phi)^{N(t)}$  in the Section A model). Assuming again pure-jump processes with bounded risk-neutral intensities, (14) implies that the continuously compounded expected index jump return is a weighted average of those of the component stocks:

$$\ln \tilde{E} J_1^\phi = \sum_{i=1}^n \phi^i \left( \ln \tilde{E} J_1^i \right) \quad (15)$$

The following proposition, which applies to both the index and the individual stocks,

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the value of  $\sigma^i$  in equation (2) is chosen such that it generates an implied volatility that matches the average implied volatility (around 40%) of our sample. Similarly, the value of  $\tilde{\lambda}$  is chosen using equation (10) where  $\hat{\sigma}'$  is again estimated from the data

<sup>36</sup>The shape of the graph does not change much when we further increase the number of stocks.

<sup>37</sup>As another example, if  $J_t^i = \prod_{j=1}^{N(t)} V_j$  where  $\{V_j; j = 1, 2, \dots\}$  are i.i.d., then  $\ln \left\{ \tilde{E}(J_1^i) \right\} = \tilde{\lambda} \tilde{E}(V_1 - 1)$ .

<sup>38</sup>See the end of appendix B for a proof.

generalizes most of the Lemma 1 results.

**Proposition 5** Fix some  $i \in \{\phi, 1, \dots, n\}$  and log return  $x \in \mathbb{R}$ , and suppose  $S^i$  satisfies (13).

a) The SDF at  $\ln(S_T/S_t) = x$  is

$$e^{r\tau} m^i(x) = \frac{\tilde{E}_t \left\{ \Phi' \left( \frac{x - \xi^i \tau - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}}{E_t \left\{ \Phi' \left( \frac{x - \xi^i \tau - \ln(J_T^i/J_t^i) - \sigma^i \rho^i \eta^c \tau}{\sigma^i \sqrt{\tau}} \right) \right\}}, \quad (16)$$

where  $\xi^i = r - \frac{1}{2}(\sigma^i)^2 - \ln \left\{ \tilde{E}(J_1^i) \right\}$ .

b) Monotonicity is characterized by

$$\frac{d}{dx} m^i(x) < 0 \quad \Longleftrightarrow \quad \frac{1}{\sigma^i} \left\{ \tilde{E}_t \left( \ln(J_T^i/J_t^i) \middle| x \right) - E_t \left( \ln(J_T^i/J_t^i) \middle| x \right) \right\} < \rho^i \eta^c \tau. \quad (17)$$

c) The sensitivity of the log-SDF slope with respect to the market price of risk is

$$\frac{d}{d(\rho^i \eta^c)} \frac{d}{dx} \ln \{ e^{r\tau} m^i(x) \} = -\frac{1}{\sigma^i} \left( 1 - \frac{1}{\tau} \text{Var}_t \left\{ \frac{\ln(J_T^i/J_t^i)}{\sigma^i} \middle| x \right\} \right). \quad (18)$$

d) Assume  $J^i$  is pure-jump process with a bounded and absolutely continuous intensity process. At the strike price  $K = e^{r\tau} S_t^i$  the slope of the implied volatility curve of asset  $i$  satisfies

$$K \frac{d}{dK} \hat{\sigma}^i(K) = \frac{\ln \tilde{E}(J_1^i)}{\sigma^i} + O(\sqrt{\tau}). \quad (19)$$

e) Let  $J$  be a Levy process, and suppose  $\ln(J_t^i) = \psi^i \ln(J_t)$ ,  $i = 1, \dots$ . Let  $\theta$  denote the market price of risk of  $\ln(J_t)$ : that is, the state-price density satisfies  $\pi_t \propto \exp(-\eta^c B_t^c - \theta \ln(J_t))$  for all  $t$ . Defining  $\delta^i$  as the difference in the per-unit-return risk compensations:

$$\delta^i = \frac{\rho^i \eta^c}{\sigma^i} - \frac{\theta}{\psi^i}, \quad (20)$$

then (7) holds and

$$m^i(x) \propto m_{BS}^i(x) E \left( J_\tau^{\delta^i \psi^i} \middle| \ln(S_\tau/S_0) = x \right). \quad (21)$$

**Proof.** See the appendix. ■

As in the special case in Section A, monotonicity depends critically on the relative contributions of the return risks and the relative market prices of risk. The violations are more likely to occur at log returns where the conditional variance of  $\ln(J_t^i/J_t^i)/\sigma^i$  is large; that is, where knowing the realized stock return leaves a lot of uncertainty about the relative contribution of the jump component of that return. Part (e) specializes the model to the case where all stocks share the same jump-risk source, and where  $\ln(J_t)$  appears linearly in the state price density (so all the components of  $\ln(J_t)$  are priced the same). Then, as in Lemma 1, stock or index  $i$ 's SDF is monotonically decreasing if  $\delta^i \leq 0$ , and violations are possible if  $\delta^i > 0$  (the proof shows that intuitive result that  $\delta^i E \left( J_\tau^{\delta^i \psi^i} \middle| \ln(S_\tau/S_0) = x \right)$  is strictly increasing in  $x$ ). Both (18) and (21) show that SDF humps are most likely when there is high uncertainty about whether jump or diffusion risk is responsible for the return, and a small change in the return  $x$  causes a large shift in the conditional jump distribution.

The general model highlights another difference between the shape of the SDF and the limiting ATM implied slope: the slope depends on only the expected jump component of return, whereas the SDF depends on the conditional jump distribution. The identity (15) shows that the index expected jump return is a weighted average of the component stock expected jump return, and is therefore invariant to the correlation structure of jumps returns across stocks.

Index SDF monotonicity violations, on the other hand, will tend to be more likely with a common jump component, as in the model (13) in which all stocks jump together. Common jumps are more likely to generate neighborhoods of index returns in which the conditional uncertainty about the relative jump component of that return is high. If the stock jumps are

all idiosyncratic (holding fixed the average expected jump  $\ln \tilde{E} \left( J_1^\phi \right)$ ), then this conditional uncertainty disappears in a large portfolio, and SDF monotonicity violations will not appear, but the implied slope will not diminish.

Together the results of the proposition show that downward jumps can be associated with SDF violations on the left side of the SDF graph (corresponding to negative asset returns), and positive jumps with SDF violations on the right side.

### III. Data

The options data are from the OptionMetrics Ivy DB database, which contains information on the entire exchange-traded US equity and index option markets and includes data on daily volume, open interest, best daily closing bid and ask quotes, option Greeks, and implied volatilities. The implied volatilities and Greeks are calculated using the American binomial model of Cox, Ross, and Rubinstein (1979). The data set also includes information on daily prices, returns and dividend distributions of all exchange-traded stocks.

In this study we consider option contracts from 1996 through 2012 that have between three and six weeks to maturity. We eliminate contracts for which the recorded ask price is lower than the bid price.<sup>39</sup> We impose a filter that requires option prices, estimated as the bid-ask midpoint, exceed the no-arbitrage lower bounds  $S - Ke^{-r\tau}$  for calls and  $Ke^{-r\tau} - S$  for puts, where  $S$  is the price of the underlying asset,  $r$  is the risk free rate, and  $\tau$  is the time to expiration. In an unreported robustness check, our results hold when we also require that option prices satisfy put-call parity bounds. That is, the bid and ask prices are required to satisfy  $C_{bid} - P_{ask} \leq S - Ke^{-r\tau} \leq C_{ask} - P_{bid}$ , where  $C$  and  $P$  are the call and put prices respectively.

Because our strike-price monotonicity results apply only to European-style options, we restrict our analysis to option contracts for which the underlying stock has no ex-dividend

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<sup>39</sup>Significant violations were observed in the data.

dates during the remaining life of the contract. The early exercise premium for such call options is therefore zero. There could be some bias in our put return results, however, because of the possibility of early exercise. In our analysis of average put option returns, this bias is small because of our short holding periods (one week) and the low interest rates within our sample period.<sup>40</sup>

Following Coval and Shumway (2001), we calculate weekly returns for the options using prices (bid-ask midpoint) observed on Tuesdays. Table I reports descriptive statistics of number and strikes and moneyness for call option contracts in the sample.<sup>41</sup> The mean (median) number of unique stocks on each buying date is 2,567 (2,477) and the mean (median) number of strikes for each stock on each buying date is 7.14 (5.00). Although the median number of strikes for each week is 5.00, Table I Panel A shows that there are 1,169,623 (729,486) unique stock  $\times$  week combinations that have at least 5 (7) strikes for each stock and 3,868 (1,874) unique stocks that over the whole sample have at least 5 (7) strikes for each week which is a fairly large sample. The total number of unique stocks for the whole sample is 6,677.

Option contracts are assigned into strike groups using the following procedure. For each stock and each week, we first identify the option strike price that is closest to the current stock price. If this strike is more than one strike-price increment away from the current stock price, then we discard all the strikes for that stock and that week.<sup>42</sup> Otherwise we assign that strike to strike group 3. The next higher strikes are assigned strike groups 4 and 5 and the two lower strikes are assigned strike groups 1 and 2. To eliminate extreme outliers, returns for each strike group are Winsorized at the 99.9% for individual stocks, and at the

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<sup>40</sup>To confirm this we estimated the early exercise premium option by option for puts using a fitted American CRR tree. The average relative premium is 0.6% for the overall sample. Furthermore, we also estimated average (adjusted) put returns by subtracting the early exercise premium from all put prices. Results are similar for the adjusted and unadjusted returns.

<sup>41</sup>The distribution for puts is very similar.

<sup>42</sup>More precisely, we follow the CBOE strike-increment rule and remove those stock-week combinations for which the absolute difference between the current stock price and nearest strike is \$2.5 or higher when current stock prices are \$25 or less, \$5 or higher when the stock price is \$25 through \$200, and \$10 or higher when the stock price is over \$200. There are 559 such cases (on average, less than one occurrence a week) which represents less than 0.1% of our final sample.

99% for the S&P 500 index.

*[Insert Table I here]*

Table I Panel B reports the mean and median levels of moneyness for each strike group in our sample. The mean (median) value of moneyness for strike group 1 is 0.768 (0.786) and 1.474 (1.277) for strike group 5. Unlike index options, five strike groups for individual stocks covers a wide range of moneyness. The thin trading in deep ITM and deep OTM options increases the possibility of stale quotes. Therefore as an (unreported) robustness test, we confirm our results with only those option contracts which have a positive volume on the buying date.

## IV. Empirical Results

Based on our Section I results, we test SDF monotonicity by testing whether option-strategy returns are increasing in strike price: calls and puts for weak monotonicity, and butterfly spreads for strict monotonicity. For each stock and each week, differences in returns across consecutive strikes are assigned to strike groups based on moneyness, and then average differences are computed for each strike group to test for positivity. By Propositions 2 and 4, strict monotonicity of the SDFs implies that each and every conditional expected return difference is strictly positive. As is typical (see for example, Coval and Shumway (2001) and Bakshi, Madan, and Panayotov (2010)), we “condition down” testing positivity of the expected difference (which is implied by positivity of the conditional expected difference).<sup>43</sup>

In Subsection A we aggregate all stocks in such moneyness bins to test overall stock SDF monotonicity. In Subsection B we similarly test monotonicity of the S&P 500. Finally, in

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<sup>43</sup>Rejection of monotonicity based on unconditional returns would imply rejection with conditional returns (a conditional expectation cannot be a.s. positive if the expectation is negative). On the other hand, with sufficient data, a stricter test of monotonicity could be designed when unconditional expectations are found to be strictly positive. However in our jump-diffusion model, with returns modeled as a Levy process, the projected SDF and the option-strategy returns do not depend on any state variables, and the “conditioning down” has no effect.

Subsection C we examine individual stock monotonicity again, but with stocks disaggregated and sorted into subsamples based on their ATM implied volatility slope, where we examine the projected SDFs of stocks that are more “index-like” in their jump-to-diffusion-risk ratio.

Our empirical strategy is to first examine the statistical significance of pairwise return differences (the t-statistics are computed, as usual, under the null of zero differences) for evidence supporting or rejecting monotonicity. If the evidence supports monotonicity, we can then apply a joint test of monotonicity, which generates a single statistic computed under the null hypothesis of nonmonotonicity (a flat SDF) to test the alternative hypothesis of a strictly increasing SDF. If the pairwise tests reject monotonicity, we apply no such joint test as there is no clear alternative hypothesis to test against. We rely on the pattern of pairwise statistics to indicate the location and extent of the violations. Given the level of noise in the returns, statistical power is an issue. This is why individual stocks are always assigned to baskets. But because the null hypothesis is always a flat SDF, a lack of power will tend to result in rejection of monotonicity in the joint test, and statistically insignificant t-statistics in the pairwise tests (again suggesting a flat SDF).

## A. Tests for Aggregated Individual Stocks

We first assign individual stock returns into strike groups using the approach outlined in Section III. For the same underlying stock, and the same time to expiration, return differences are calculated by subtracting the option strategy return for a particular strike from the option strategy return for the immediate higher strike.<sup>44</sup> Average return differences are reported in Table II with, for example, column “3 – 2” presenting average strike-group-3 minus strike-group-2 returns. For a given strategy (e.g., long calls), each difference used to compute the average is obtained by subtracting the strike-group-2 from strike-group-3 returns for the same stock on the same week. Under the hypothesis of weak monotonicity, each call

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<sup>44</sup>If a particular option price is missing or does not satisfy the no-arbitrage restriction, then the associated return differences are considered as missing. For example if the strike group 2 return for a stock on a particular week is missing, then we assign the return differences “2 – 1” and “3 – 2” as missing.

and put return difference has a positive expected value, and under the hypothesis of strict monotonicity, all strategy return differences have positive expected values. The return differences are averaged across different stocks to form a weekly time series of average return differences. The average of the weekly time series and the corresponding t-statistics are reported in Table II.

*[Insert Table II here]*

For vanilla calls and puts we compute an additional difference. For vanilla call options, the column “ $1 - R_S$ ” presents the average difference between each strike-group-1 return and the return of the underlying stock (which represents a zero-strike call). For put options, the column “ $R_f - 5$ ” presents the average difference between the risk-free rate (the limiting return of a put as the strike price increases towards infinity) and the strike-group-5 return.

Table II also provides a skewness-adjusted t-statistic for the average return difference. Central-limit approximations are problematic because of the small sample sizes and the heavy skewness of option returns (deep OTM option returns could be close to -100%). We use skewness-adjusted standard errors developed by Chen (1995) as an extension of Johnson (1978) and Sutton (1993). It is derived using Hall’s t-type inversion of the Edgeworth expansion and has been shown to be accurate for sample sizes as small as 13 and for distributions as asymmetric as the exponential distribution. Further details on this test statistic are provided in the appendix.

### **A.1. Weak Monotonicity Tests**

Results for call option returns, presented in Table II Panel A, indicate that the average return differences are all positive (return differences “5 – 4”, “4 – 3” and “3 – 2” are each significant at the 1% level) consistent with expected call returns monotonically increasing in the strike price. For example, call options in strike group 1 earn on average 0.2% higher weekly return than that of underlying stocks, and call options in strike group 4 earn about 6.7% higher

average weekly return than call options in strike group 3. The skewness-adjusted t-statistic for OTM contracts are substantially higher than the usual t-statistics reflecting the strong positive skew in OTM call return distributions.<sup>45</sup>

The put-option results in Panel B are somewhat weaker. Only two out of five of the return differences are significant at the 5% level. The average “3 – 2” return difference is 0.9% and the average “2 – 1” return difference is 0.4%. However in an unreported test, we confirm that for both the call and put options the average return difference “5 – 1” is positive and significant at the 5% level of significance. Moreover, the Page-test result in Section A.3 is consistent with strict monotonicity.

## A.2. Strict Monotonicity Tests

Butterfly spread returns are used to test for strict monotonicity.<sup>46</sup> A symmetric butterfly spread with call options is composed of two short calls with strike price  $K$ , and long one call each with strikes  $K - \Delta K$  and  $K + \Delta K$ .<sup>47</sup> The butterfly spread therefore has a payoff only within the stock-price window defined by the lower and upper strikes. Because its expected return is inversely related to a weighted average of the SDF within this window. As the strike is shifted to the right, so is the window, and a decreasing  $m_{t,T}()$  implies higher expected returns.

To calculate the return differences for the butterfly spreads, we impose the following restrictions: 1) There should exist at least three valid strikes for any underlying stock on any buying date to estimate at least one return difference; 2) The estimated price of the

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<sup>45</sup>This is not surprising because of the following reason: it is well known that the distribution of the t-statistic inherits a skewness that is opposite of the parent distribution. If the original distribution is positively skewed, then the t-distribution is negatively skewed, and confidence intervals are to the left of where they should be to ensure symmetric coverage. In the context of our hypothesis, testing this translates into true upper tail probabilities that are higher than normal resulting in fewer rejections. Thus to find a more accurate and powerful test with a common critical point for a given significance level  $\alpha$ , we use a modified t-statistic that appropriately accounts for the skewness.

<sup>46</sup>There are other option strategies belonging to the log-concave payout class that potentially contain information about the SDF. Particularly relevant among these are binary options as the CBOE already introduced binary options on SPX and VIX indexes in July 2008.

<sup>47</sup>Butterfly spread strike groups are assigned based on the middle strike.

spread must be greater than \$0.01 to eliminate no-arbitrage convexity violations and reduce the possibility of unrealistic returns; and 3) The value of  $\Delta K$  should be the same across all butterfly spreads on the same underlying stock and the same buying date. We allow the spreads to have overlapping strikes: that is, the middle and right strike of a spread could be the left and middle strike respectively of the next higher spread. Results from Table II Panel C indicate that, consistent with strict monotonicity, all the return differences are positive and significant at the 1% level.

### A.3. Joint Monotonicity Tests

We apply two joint tests of monotonicity:<sup>48</sup> a modification of the Page test for ordered alternatives, and a bootstrapping test introduced by Patton and Timmermann (2010). The Page test is implemented by first computing, for each period, the average return by strike group, and then the rank of this vector of average returns. For example, the rank of the vector will be  $(1, \dots, k)$  for  $k$  strike groups if the average returns are monotonically increasing that period. The vector of ranks for each period are then averaged over time. The vector of average ranks is regressed against the strike group number  $(1, \dots, k)$ . The Page statistic can be interpreted as the ratio of the regression slope to its standard error. If average returns tend to increase with strike (regardless of the magnitude of the increments), then the average ranks and strike-group number will tend to be highly correlated, yielding a large statistic and a rejection of the null (no linear relationship between average rank and strike-group number) in favor of the alternative hypothesis of increasing average returns across strikes.

The test of Patton and Timmermann (2010) uses bootstrapping to estimate the distribution of the minimum average return increment. If the minimum average increment is sufficiently large relative to the estimated empirical distribution of the minimum computed under the null, then we reject the null of equal increments (the null hypothesis least favorable to the alternative) in favor of the alternative hypothesis of average returns strictly increasing

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<sup>48</sup>See Appendix D for a more detailed description of the joint tests and for our modification of the Page test.

with strike. The bootstrapping test could be viewed as stronger because every increment in average returns must be sufficiently large to be consistent with the alternative hypothesis. However, it could be more influenced by outliers than the rank-based Page test.

We emphasize that in both joint tests, the null hypothesis is nonmonotonicity (zero slope for the Page test, and equal expected returns under the bootstrapping test). Therefore the absence of statistical power (if, for example, we apply the test to a single stock rather than a basket), will tend to result in the rejection of strict monotonicity.

Table III shows that the Page statistics for all the option strategies are positive and significant at the 1% level, consistent with the alternative hypothesis that the expected returns of the option strategies are monotonically increasing across strike groups. For example, the statistic for call (butterfly) strategy is 22.571 (14.869) for the classical Page test and 19.523 (8.360) for the modified Page test.

*[Insert Table III here]*

Table III also reports the results from the Patton and Timmermann (2010) monotonicity test. We report the  $p$ -values for both the studentized and non-studentized versions. The studentized  $p$ -values are calculated by scaling the estimates from the bootstrap samples by the corresponding standard errors of the samples. The non-studentized (studentized)  $p$ -values for call and butterfly strategies are 0.098 (0.003) and 0.092 (0.001) respectively. The results from both joint tests therefore provide compelling evidence supporting strict monotonicity.

## **B. Testing S&P 500 SDF Monotonicity**

Subsection A provides strong evidence that the typical individual-stock SDF is monotonically decreasing. On the other hand, Section II shows that in a jump-diffusion model diversification of Brownian risk can result in SDF humps even if the SDF every component stock is monotonic. We apply our monotonicity tests to the S&P 500 and, consistent with

the preponderance of the empirical literature on S&P 500 index monotonicity, find evidence of violations.

*[Insert Table IV here]*

Table IV shows that average call returns are decreasing in strike for low (ITM) strikes (statistically significant at the 5% level for the second strike difference), and butterfly returns are decreasing in strike for the low strikes. These results indicate hump on the left side of the index SDF graph, and are consistent with our Section II in which the typical stock exhibits downward jump risk (as evidenced by the downward sloping implied smile of the typical stock), but the jump risk relative to Brownian risk is insufficient at the individual stock level to induce an SDF hump.

### C. Disaggregated Individual Stocks

Although our Subsection A empirical results indicate that the average stock has a downward sloping SDF, we investigate if stocks that have more “index-like” properties share similar patterns of SDF monotonicity violations. Section II shows the key role of the jump-return distribution relative to Brownian diffusion in generating SDF humps, suggesting that the normalized implied smile slope, which Proposition 5 shows measures the ratio of the expected jump to Brownian diffusion, would be a good predictor of SDF humps. As in equation (10), the implied slope is normalized by multiplying the slope with the ATM strike. Our measure of slope is slightly different from what is usually used in the literature: we denote this alternative measure  $\Pi = K\hat{\sigma}'(K)/\hat{\sigma}(K)$ .<sup>49</sup> From Table V we see that the average normalized slopes are -0.181 and -0.649 for individual stocks and the S&P 500, respectively. The Table shows that dividing by  $\hat{\sigma}(K)$  (which tends to be smaller for the index) amplifies both the level and the relative differences under the slope-measure  $\Pi$ .

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<sup>49</sup>For example, Bakshi, Kapadia, and Madan (2003), among others, calculates the slope by estimating the equation  $\ln(\hat{\sigma}(y)) = \Pi_0 + \Pi \ln(y) + \epsilon$ , where  $y$  is the moneyness of the option. This yields the at-the-money slope  $\Pi = K \frac{\hat{\sigma}'(K)}{\hat{\sigma}(K)}$ .

*[Insert Table V here]*

We sort stocks into quintiles according to the normalized slope of ATM implied smile.<sup>50</sup> For example, a stock with a large negative implied slope would be associated with stronger downside jump potential relative to diffusion risk and would be assigned to the lowest quintile.

The lowest quintile stocks are going to be more “index-like” with regards to their relative contribution of jumps to returns and are more likely to demonstrate similar patterns of monotonicity violations. Table V shows that the average normalized slope of the lowest quintile is similar to that of the index: -0.721 versus -0.649, respectively.

*[Insert Table VI here]*

*[Insert Table VII here]*

Tables VI, VII, and VIII repeat the analysis in Table II by quintiles of the implied slope for call, put, and butterfly call spreads respectively. We find that a pattern of SDF monotonicity violations emerges that is similar to that of the S&P 500 index. For call and butterfly spreads we find severe violations of SDF monotonicity corresponding to negative returns in the bottom quintile. Interestingly, the Table VIII butterfly spread results for the top quintile reveal an SDF hump corresponding to positive returns (the put results in VII suggest an additional possible hump far to the left, below the strike range captured by the butterfly spreads), which is consistent with the upside jump potential of this quintile (Table V shows that the top quintile has a positive implied slope).

*[Insert Table VIII here]*

We conclude our empirical section by summarizing the characteristics of stocks by quintiles of implied volatility slope. This allows us to investigate whether stocks in the bottom

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<sup>50</sup>The ATM slope for each stock is calculated by finding the nearest ATM options, and computing the ratio of the implied volatility difference to the strike difference. Our results insensitive to the normalization. We get very similar results if we sort based on  $\Pi$  instead.

quintile look more “index-like” along other dimensions besides implied slope. Table IX shows that stocks in the bottom quintile have market betas close to 1, and, compared to the other quintiles, slightly higher  $R^2$ s, and slightly lower  $RMSE$ s. The highest quintile stocks (which shows evidence of positive jumps), have smaller market values, lower betas and  $R^2$ s, and higher  $RMSE$ s.<sup>51</sup>

*[Insert Table IX here]*

We have also explored other sorting criteria and other subsamples based on additional characteristics of the underlying stock. These characteristics include size, market-to-book ratios, liquidity, and whether the firm is a financial or not.<sup>52</sup> Sorting on these characteristics resulted in no patterns of SDF monotonicity violations.

## V. Conclusion

We show that a simple jump-diffusion model can explain humps (or, local violations of monotonicity) in the graph of a projected SDF. Humps in the projected SDF are caused by a discrepancy in the per-unit-return pricing of diffusion and jump risk, and they are most likely to appear in the vicinity of returns where uncertainty about whether the return is caused by jumps or diffusion risk is high. If stocks share a large common risk factor driving jumps, then diversification amplifies this discrepancy, resulting in an index SDF hump despite monotonicity in the typical stock. The same model predicts a steeper normalized implied volatility curve for the index than the typical stock. Our empirical evidence is consistent with the model. We show that in contrast to the index, which exhibits a hump on the left side of the SDF graph, the average stock has a downward sloping projected SDF.

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<sup>51</sup>The betas,  $R^2$ , and  $RMSE$  are estimated for each underlying stock and each month from the Fama-French and Carhart 4-factor regression model using daily returns. Similar results are obtained using just the CAPM or Fama-French 3 factor model

<sup>52</sup>Stocks with market capitalization less than \$1 billion are classified as small cap, \$1 billion to \$5 billion as mid-cap, and greater than \$5 billion as large cap. Market-to-book ratios are calculated using definitions provided on Kenneth French’s website, and we use trading volume of the underlying stock on the buying date of the option strategy as a proxy for liquidity. Firms are classified as financials if their SIC codes are between 6000 and 6999.

Our model also predicts that stocks exhibiting stronger downside (upside) jump potential relative to diffusion risk are more likely to exhibit monotonicity violations corresponding to negative (positive) returns. We show that the slope of the implied volatility smile provides a measure of jump-to-diffusion risk (in a general jump-diffusion model). Consistent with this theory, our empirical results find SDF violations corresponding to negative stock returns for stocks that have a steeply negative smile slope, and violations corresponding to positive returns for stocks that have a positive smile slope. We also show that stocks which exhibit left-side SDF monotonicity violations share some characteristics with the S&P 500 index: implied smile slopes similar to S&P 500, market betas close to one, and slightly lower idiosyncratic volatility (compared to the other quintiles sorted on slope). Stocks with right-side SDF violations, on the other hand, tend to have smaller market value, lower betas, and higher idiosyncratic volatility.

We devise a test of SDF monotonicity based on option-trading strategy returns that is model-free and avoids the need to estimate the SDF. We show that the SDF is strictly monotonic if and only if (conditional) expected returns of a class of option trading strategies are increasing in the strike price. Our test of SDF monotonicity therefore reduces to the relatively simpler problem of testing whether expected returns of a class of trading strategies are increasing in strike price. Furthermore, this problem is well suited to tests of overall monotonicity. We apply both a modification to the Page test for ordered alternatives, as well as a recent bootstrapping-based test of overall monotonicity introduced in Patton and Timmermann (2010). Our empirical results support strict SDF monotonicity for the average stock.

While we have focused on testing monotonicity of the SDF, option-trading-strategy returns can potentially be used to test other characteristics of the SDF shape, such as convexity (see, for example, Bakshi, Madan, and Panayotov (2010) who test for U-shapes in the right tail of the SDF). The Page test can also be modified to test other patterned alternatives. Furthermore, our results characterizing monotonicity of the projected SDF in terms

of option-trading strategy returns is applicable to any underlying process (whether or not a traded asset) on which options are traded, such as energy or commodity products, or the VIX.

## Appendix A SDF monotonicity

Recall that throughout this paper, we have fixed a time period  $[t, T]$  and use time- $t$  (conditional) expectations and covariances. For simplicity of notation, in the appendix we omit all time- $t$  subscripts and use time-0 expectations and covariances. This is without loss of generality because we are not assuming a trivial (uninformative) time-0 filtration. All of our results apply with time- $t$  conditional expectations and covariances, time  $[t, t + T]$  returns, and the (conditional) SDF for the time interval  $[t, t + T]$ .

The proofs of Propositions 3 and 4 below use the following Lemma.

**Lemma A1** *Let  $(\Omega, \mathcal{F}, P)$  be a probability space, let  $\mathcal{B}$  denote the collection of Borel sets of real numbers, and let  $Y$  be a random variable with distribution function  $\Phi$ . Assume  $E|Yf(Y)|$ ,  $E|Y|$  and  $E|f(Y)|$  are all finite.*

*a) If  $f$  is a strictly decreasing function of finite variation then*

$$\text{Cov}(Y, f(Y) | Y \in B) < 0 \quad \text{for all } B \in \mathcal{B} \text{ with } P(Y \in B) > 0.$$

*b) If  $f$  is differentiable, and  $\Phi$  is strictly increasing on  $[a, b]$  (no zero probability intervals), then  $f$  is strictly decreasing on  $[a, b]$  if and only if*

$$\text{Cov}(Y, f(Y) | Y \in [\alpha, \beta]) < 0 \quad \text{for all } [\alpha, \beta] \in [a, b]. \quad (\text{A1})$$

**Proof.** Part a) Consider first the case when  $|f(Y)|$  is bounded. Define

$$h(x; B) = \int_{-\infty}^x \{E(Y | B) - y\} 1_{\{y \in B\}} \frac{d\Phi(y)}{P(Y \in B)},$$

which satisfies  $h(-\infty) = h(\infty) = 0$  and  $h(x) > 0$  for all  $x$  such that  $P(Y \in B \text{ and } Y \leq x) >$

0. Using integration by parts:

$$\begin{aligned}\text{Cov}(Y, f(Y) | Y \in B) &= \int_{-\infty}^{\infty} (y - E(Y | B)) 1_{\{y \in B\}} f(y) \frac{d\Phi(y)}{P(Y \in B)} \\ &= - \int_{-\infty}^{\infty} f(y) dh(y; B) \\ &= \int_{-\infty}^{\infty} h(y-; B) df(y).\end{aligned}$$

If  $f$  is strictly decreasing then the left hand side must be strictly negative. The case of unbounded  $|f(Y)|$  is proven using the dominated convergence theorem.

Part b) The sufficiency of  $f$  strictly decreasing follows from Part a. To prove necessity, fix some  $\alpha \in [a, b]$  and define, for all  $\varepsilon > 0$ ,

$$\Lambda(\varepsilon) = \sup_{z \in [0, \varepsilon]} \left| \frac{f(\alpha + z) - f(\alpha)}{z} - f'(\alpha) \right|$$

and note that differentiability of  $f$  at  $\alpha$  is equivalent to  $\lim_{\varepsilon \downarrow 0} \Lambda(\varepsilon) = 0$ . Define  $v(Y - \alpha)$  implicitly by

$$f(Y) = f(\alpha) + (Y - \alpha) f'(\alpha) + (Y - \alpha) v(Y - \alpha) \quad (\text{A2})$$

and note that  $|v(Y - \alpha)| \leq \Lambda(\varepsilon)$  for  $Y \in [\alpha, \alpha + \varepsilon]$ . Substituting into (A1) (with  $\beta = \alpha + \varepsilon$ ) we get

$$0 > \frac{\text{Cov}(Y, f(Y) | Y \in [\alpha, \alpha + \varepsilon])}{\text{Var}(Y | Y \in [\alpha, \alpha + \varepsilon])} = f'(\alpha) + \frac{\text{Cov}(Y, (Y - \alpha) v(Y - \alpha) | Y \in [\alpha, \alpha + \varepsilon])}{\text{Var}(Y | Y \in [\alpha, \alpha + \varepsilon])}.$$

Applying Cauchy-Schwartz together with  $\text{Var}((Y - \alpha) v(Y - \alpha) | Y \in [\alpha, \alpha + \varepsilon]) \leq \varepsilon^2 \Lambda(\varepsilon)^2$  implies

$$\left| \frac{\text{Cov}(Y, (Y - \alpha) v(Y - \alpha) | Y \in [\alpha, \alpha + \varepsilon])}{\text{Var}(Y | Y \in [\alpha, \alpha + \varepsilon])} \right| \leq \Lambda(\varepsilon).$$

Taking the limit as  $\varepsilon \downarrow 0$  we get  $f'(\alpha) \leq 0$  for every  $\alpha \in [a, b]$ . Therefore  $f$  must be nonincreasing on  $[a, b]$ . Finally, suppose  $f$  is not strictly decreasing on  $[a, b]$ . Then there must exist points  $\{c, d\}$  satisfying  $a \leq c < d \leq b$  such that  $f(c) = f(d)$ . But this implies

$f'(x) = 0$  for  $x \in [c, d]$ , contradicting (A1). ■

**Remark A1** *It is easy to extend part (b) by allowing  $f$  to have a finite number of jumps, and requiring differentiability only between jumps. Then we can rule out any upward jump in  $f$  by using (A1) to evaluate the covariance term for a sufficiently small interval around the jump (because  $\int_{-\infty}^{\infty} h(y-; B) df(y)$  will be dominated by the positive jump term).*

### Proof of Proposition 1

(i)  $\iff$  (ii) follows because

$$R(S_T) = \frac{E(S_T)}{E(mS_T)} > \frac{1}{E(m)} = R(1) \quad (\text{A3})$$

is equivalent to  $0 > E(mS_T) - E(S_T)E(m)$ , which is equivalent to (i). Rearranging (A3) gives  $E(S_T) > E(mS_T)/E(m)$ , which is equivalent to (iii).

### Proof of Proposition 2

(i)  $\iff$  (iv) follows, as shown in Coval and Shumway (2001), from

$$\begin{aligned} \frac{d}{dK} R((S_T - K)^+) &= \frac{d}{dK} \frac{E\{(S_T - K)^+\}}{E\{m(S_T - K)^+\}} \\ &= \frac{E\{(S_T - K)^+\} E(m1_{\{S_T > K\}}) - E(m(S_T - K)) P(S_T > K)}{(E\{m(S_T - K)^+\})^2} \\ &= - \left( \frac{P(S_T > K)}{E\{m(S_T - K)^+\}} \right)^2 \text{Cov}(S_T, m \mid S_T > K) \end{aligned}$$

and, similarly,

$$\frac{d}{dK} R((K - S_T)^+) = - \left( \frac{P(S_T < K)}{E\{m(K - S_T)^+\}} \right)^2 \text{Cov}(S_T, m \mid S_T < K).$$

(i)  $\iff$  (ii) follows because

$$R(S_T 1_{\{S_T \in [\alpha, \beta]\}}) = \frac{E(S_T 1_{\{S_T \in [\alpha, \beta]\}})}{E(m S_T 1_{\{S_T \in [\alpha, \beta]\}})} > \frac{P(S_T \in [\alpha, \beta])}{E(m 1_{\{S_T \in [\alpha, \beta]\}})} = R(1_{\{S_T \in [\alpha, \beta]\}})$$

is equivalent to

$$0 > E(m S_T 1_{\{S_T \in [\alpha, \beta]\}}) P(S_T \in [\alpha, \beta]) - E(S_T 1_{\{S_T \in [\alpha, \beta]\}}) E(m 1_{\{S_T \in [\alpha, \beta]\}}). \quad (A4)$$

The right-hand side of (A4) is equal to  $Cov(S_T, m \mid S_T \in [\alpha, \beta]) \{P(S_T \in [\alpha, \beta])\}^2$ . The inequality in (iii) is

$$E(S_T \mid S_T \in [\alpha, \beta]) = \frac{E(m S_T 1_{\{S_T \in [\alpha, \beta]\}})}{P(S_T \in [\alpha, \beta])} > \frac{E(m S_T 1_{\{S_T \in [\alpha, \beta]\}})}{E(m 1_{\{S_T \in [\alpha, \beta]\}})} = E^Q(S_T \mid S_T \in [\alpha, \beta]).$$

Together with (A4) we get (i)  $\iff$  (iii).

### Derivation of Example 1

Let  $m_i = m(s_i)$  for  $i \in \{1, \dots, 4\}$ . We derive necessary and sufficient conditions on  $m$  for weak monotonicity. For the truncated regressions it is sufficient to consider conditioning on  $\{S_T \leq s_i\}$  or  $\{S_T \geq s_i\}$ , for  $i \in \{2, 3\}$ . Obviously (as in a two-point regression)

$$\begin{aligned} \text{Cov}(S_T, m \mid S_T \leq s_2) &< 0 \iff m_1 > m_2 \\ \text{and } \text{Cov}(S_T, m \mid S_T \geq s_3) &< 0 \iff m_3 > m_4. \end{aligned}$$

Noting that  $E(S_T \mid S_T \leq s_3) = s_2$  (recall that the  $s_i$  are equally spaced):

$$\begin{aligned} \text{Cov}(S_T, m \mid S_T \leq s_3) &= E((S_T - s_2) m \mid S_T \leq s_3) \\ &= \frac{1}{3} \{(s_1 - s_2) m_1 + (s_2 - s_2) m_2 + (s_3 - s_2) m_3\} \end{aligned}$$

and therefore

$$\text{Cov}(S_T, m \mid S_T \leq s_3) < 0 \iff m_1 > m_3.$$

The analogous argument gives

$$\text{Cov}(S_T, m \mid S_T \geq s_2) < 0 \iff m_2 > m_4.$$

Finally, the unconditional covariance satisfies (using  $E(S_T) = (s_2 + s_3)/2 = \bar{s}$ )

$$\text{Cov}(S_T, m) = E((S_T - \bar{s})m) = \frac{1}{4} \{(s_1 - \bar{s})m_1 + (s_2 - \bar{s})m_2 + (s_3 - \bar{s})m_3 + (s_4 - \bar{s})m_4\}$$

and therefore (multiplying both sides by 8 and defining  $\Delta s = s_{i+1} - s_i$  as the common stock-price increment)

$$\text{Cov}(S_T, m) < 0 \iff \Delta s \cdot m_3 + 3\Delta s \cdot m_4 < 3\Delta s \cdot m_1 + \Delta s \cdot m_2.$$

That is, negativity of the unconditional covariance is equivalent to

$$m_3 - m_2 < 3(m_1 - m_4)q; \tag{A5}$$

but (A5) follows trivially from the earlier conditions  $m_1 > m_3$  and  $m_2 > m_4$  (which together imply  $m_3 - m_2 < m_1 - m_4$ ).

### Proof of Proposition 3

Let  $g$  be some regular payout function which we assume, for simplicity, to be right continuous at  $x_1$  and left continuous at  $x_2$ ; that is, if  $x_1 > -\infty$  then  $g(x_1) = g(x_1+)$  ( $= \lim_{s \downarrow x_1} g(s)$ ) and if  $x_2 < \infty$  then  $g(x_2) = g(x_2-) (= \lim_{s \uparrow x_2} g(s))$ . The distribution function of  $S_T$  is denoted  $F()$ , which is assumed to have a density function which is strictly positive on  $(0, \infty)$ . Finally, define  $\Delta g(x_i) = g(x_i+) - g(x_i-)$ ,  $i \in \{1, 2\}$  as the jump in the

payoff function at  $x_i$  if  $x_i$  is finite, and zero otherwise.

a) Sufficiency of (1): Define the change of measure

$$\frac{dQ^K}{dP} = \frac{g(S_T - K)}{E(g(S_T - K))},$$

and let  $E^{Q^K}$  denote expectation under the measure  $Q^K$ . Consider first the case when  $g$  is continuous ( $\Delta g(x_1) = \Delta g(x_2) = 0$ ). We differentiate the inverse of the expected return,

$$\frac{1}{R(g(S_T - K))} = \frac{E(mg(S_T - K))}{E(g(S_T - K))} \quad (A6)$$

to get

$$\begin{aligned} & \frac{d}{dK} \frac{1}{R(g(S_T - K))} \\ = & \frac{E(m(S_T)g(S_T - K))E(g'(S_T - K)) - E(g(S_T - K))E(m(S_T)g'(S_T - K))}{\{E(g(S_T - K))\}^2} \end{aligned}$$

and therefore

$$\begin{aligned} \frac{d}{dK} \frac{1}{R(g(S_T - K))} &= E^{Q^K}(m(S_T))E^{Q^K}\left(\frac{g'(S_T - K)}{g(S_T - K)}\right) - E^{Q^K}\left(m(S_T)\frac{g'(S_T - K)}{g(S_T - K)}\right) \\ &= -\text{Cov}^{Q^K}\left(m(S_T), \frac{g'(S_T - K)}{g(S_T - K)}\right). \end{aligned}$$

By Lemma A1, and the assumption that  $m(\cdot)$  is monotonically decreasing,  $g'(x)/g(x)$  non-increasing (decreasing) implies that  $R(g(S_T - K))$  is nondecreasing (increasing)  $K$ .<sup>53</sup>

Now consider the case when  $g$  has discontinuities. Because  $g$  is regular, discontinuities can only occur at  $x_1$  if  $x_1 > -\infty$  or at  $x_2$  if  $x_2 < \infty$ . Differentiating (A6) with discontinuous  $g$  introduces additional terms at the jumps of  $g$ . For example,

$$\frac{d}{dK} E(m(S_T)g(S_T - K)) = -E(mg'(S_T - K)) - \sum_{i=1}^2 \Delta g(x_i) m(K + x_i) F'(K + x_i)$$

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<sup>53</sup>To apply the Lemma, define  $Y = m(S_T)$ , and the decreasing function  $f(Y) = \frac{g'}{g}(m^{-1}(Y))$ .

(note that  $g'$  exists a.e., with the possible exception of the points  $S_T \in \{K + x_1, K + x_2\}$ ) with an analogous expression for  $\frac{d}{dK} E(g(S_T - K))$ . This yields

$$\begin{aligned} \frac{d}{dK} \frac{1}{R(g(S_T - K))} &= -\text{Cov}^{Q^K} \left( m(S_T), \frac{g'(S_T - K)}{g(S_T - K)} \right) \\ &\quad + \sum_{i=1}^2 \Delta g(x_i) F'(K + x_i) \frac{E^{Q^K} \{m(S_T) - m(K + x_i)\}}{E(g(S_T - K))}. \end{aligned} \quad (A7)$$

From  $\Delta g(x_1) \geq 0$  and  $\Delta g(x_2) \leq 0$ , and  $Q^K(S_T \in (K + x_1, K + x_2)) = 1$  (all the probability mass under  $Q^K$  is concentrated where the payoff function is positive), then  $m$  decreasing implies that the discontinuity terms on the right side of (A7) are nonpositive.

b) Necessity of (1): We sketch the necessity proof, which follows easily from (A7). Suppose contrary to the hypothesis that

$$\frac{g'(s_2)}{g(s_2)} > \frac{g'(s_1)}{g(s_1)}$$

for some  $s_1 < s_2$ , where  $s_1, s_2 \in (x_1, x_2)$ . Then we can construct a distribution function  $F$  which is sufficiently concentrated around the two prices  $S_T \in \{s_1 + K, s_2 + K\}$  (recall that regularity implies piecewise continuity of  $g'(x)/g(x)$  for  $x \in (x_1, x_2)$ ) so that (A7) implies that  $\frac{d}{dK} \{R(g(S_T - K))\}^{-1} > 0$  (note that concentrating the distribution around these two points also implies that the discontinuity terms on the right side of (A7) become negligible).

**Remark A2**  $R(g(S_T - K))$  is strictly increasing in  $K$  if either  $g'(x)/g(x)$  is nonincreasing (on  $(x_1, x_2)$ ) and  $g$  is not continuous, or if  $g'(x)/g(x)$  is nonincreasing and nonconstant and  $g$  is continuous. This follows from (A7) and the assumptions that  $m(\cdot)$  is strictly decreasing and that the density of  $F$  is strictly positive on  $(0, \infty)$ . That is, we need either a discontinuity in  $g$  or a strictly decreasing part of  $g'(x)/g(x)$  for  $x \in (x_1, x_2)$  to obtain strict monotonicity of  $R(g(S_T - K))$  in  $K$ . The exponential class,  $g(x) = \exp(\alpha + \beta x)$ , for some  $\alpha, \beta \in \mathbb{R}$  ( $x_1 = -\infty$  and  $x_2 = \infty$ ), is the unique class of regular payoff functions for which

$R(g(S_T - K))$  is invariant to changes in  $K$ . Note that it is continuous and  $g'(x)/g(x)$  is constant. On the other hand,  $g(x) = 1_{\{x \geq 0\}}$ , the payoff of a binary cash or nothing call ( $x_1 = 0$  and  $x_2 = \infty$ ) satisfies  $g'(x)/g(x) = 0$  for  $x \in (x_1, x_2)$ , yet  $R(g(S_T - K))$  is strictly decreasing in  $K$  because  $\Delta g(0) = 1$ .

$R(g(S_T - K))$  is strictly increasing in  $K$  under the condition of the proposition unless the payoff function is within the exponential class:  $g(x) = \exp(\alpha + \beta x)$ , for some  $\alpha, \beta \in \mathbb{R}$ . Within this class  $R(g(S_T - K))$  is invariant to  $K$ .

We conclude by justifying the statement made in the paper that an increase in strike causes a first-order-stochastic-dominance shift in the stock-price distribution function under  $Q^K$ . This can be also used as an alternative proof to (a) above. The distribution function of  $S_T$  under  $Q^K$  is

$$F^{Q^K}(s) = Q^K(S_T \leq s) = \frac{E[g(S_T - K) 1_{\{S_T \leq s\}}]}{E[g(S_T - K)]} \quad s \in [0, \infty). \quad (\text{A8})$$

We show that  $F^{Q^{K_2}}(\cdot)$  stochastically dominates  $F^{Q^{K_1}}(\cdot)$  in the first order sense for any  $0 \leq K_1 < K_2$ .<sup>54</sup> We consider only the case where  $g$  is continuous (the modifications for discontinuities are similar to those in part (a) above). Differentiating (A8) we get  $\frac{d}{dK} F^{Q^K}(s) = 0$  for any  $s < x_1 + K$  and

$$\frac{d}{dK} F^{Q^K}(s) \leq 0 \quad \Longleftrightarrow \quad \frac{E[g'(S_T - K) 1_{\{S_T \leq s\}}]}{E[g(S_T - K) 1_{\{S_T \leq s\}}]} \geq \frac{E[g'(S_T - K)]}{E[g(S_T - K)]} \quad \text{all } s > x_1 + K.$$

Applying the change of measure, this is equivalent to

$$\frac{d}{dK} F^{Q^K}(s) \leq 0 \quad \Longleftrightarrow \quad E^{Q^K} \left[ \frac{g'(S_T - K)}{g(S_T - K)} \mid S_T \leq s \right] \geq E^{Q^K} \left[ \frac{g'(S_T - K)}{g(S_T - K)} \right]$$

for all  $s \geq x_1 + K$ , which is clearly satisfied if  $g'/g$  is nonincreasing.

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<sup>54</sup>See Huang and Litzenberger (1988)

## Proof of Proposition 4

The proof of (i)  $\iff$  (ii) follows from Lemma A1 above. The proof of the equivalence of (ii), (iii) and (iv) is the essentially the same as in the proof of Proposition 2. The proof of (i)  $\implies$  (v) follows from Proposition 3. The proof of (v)  $\implies$  (i) follows from the limiting result at the end of Example 3.

## Appendix B Jump Diffusion Models

Some of the Proposition 5 results are proven using the following lemma, which gives an expression for the SDF and a necessary and sufficient condition for monotonicity when prices are lognormal conditional under both  $P$  and  $\tilde{P}$  after conditioning on a (possibly multidimensional) state variable process  $J = \{J_t; t \in [0, T]\}$ . We consider only the case of a jump process, but  $J$  could instead represent a Markov regime or stochastic volatility.

**Lemma B2** *Fix some  $t < T$  and  $i \in \{\phi, 1, 2, \dots, n\}$ , and suppose that conditional on the state-variable process  $J$  the price  $S_T^i$  is lognormally distributed under both  $P$  and  $\tilde{P}$  with*

$$E(\ln(S_T^i/S_t^i) | J, \mathcal{F}_t) = \alpha(J), \quad \tilde{E}(\ln(S_T^i/S_t^i) | J, \mathcal{F}_t) = \tilde{\alpha}(J), \quad (\text{B9a})$$

$$\text{Var}(\ln(S_T^i/S_t^i) | J, \mathcal{F}_t) = \widetilde{\text{Var}}(\ln(S_T^i/S_t^i) | J, \mathcal{F}_t) = v(J), \quad (\text{B9b})$$

where  $\alpha(J)$ ,  $\tilde{\alpha}(J)$ , and  $v(J)$  are assumed constant conditional on  $J$ . Then

$$m^i(x) = e^{-r(T-t)} \frac{\tilde{E}_t \left\{ \frac{1}{\sqrt{v(J)}} \Phi' \left( \frac{x - \tilde{\alpha}(J)}{\sqrt{v(J)}} \right) \right\}}{E_t \left\{ \frac{1}{\sqrt{v(J)}} \Phi' \left( \frac{x - \alpha(J)}{\sqrt{v(J)}} \right) \right\}}, \quad \text{all } s > 0. \quad (\text{B10})$$

Differentiating we get

$$\frac{d}{dx} m^i(x) < 0 \quad \iff \quad \tilde{E}_t \left( \frac{x - \tilde{\alpha}(J)}{v(J)} \middle| x \right) > E_t \left( \frac{x - \alpha(J)}{v(J)} \middle| x \right), \quad (\text{B11})$$

for any  $x \in \mathbb{R}$ .

**Proof.** We omit the superscript  $i$  throughout. The objective conditional distribution function is

$$P(\ln(S_T/S_t) \leq x | J, \mathcal{F}_t) = \Phi\left(\frac{x - \alpha(J)}{\sqrt{v(J)}}\right).$$

Differentiating with respect to  $s$  and taking expectations gives the density

$$\frac{d}{ds}P(\ln(S_T/S_t) \leq x | \mathcal{F}_t) = E_t\left\{\frac{1}{\sqrt{v(J)}}\Phi'\left(\frac{x - \alpha(J)}{\sqrt{v(J)}}\right)\right\}.$$

The derivation of the discounted risk-neutral density is analogous. The SDF is therefore (B10). Differentiating with respect to  $x$  implies that  $\frac{d}{dx}m(x) < 0$  if and only if (omitting  $J$  arguments)

$$\tilde{E}_t\left\{\frac{e^{-r_t, T(J)}\frac{1}{\sqrt{v}}\Phi'\left(\frac{x-\tilde{\alpha}}{\sqrt{v}}\right)}{\tilde{E}_t\left\{\frac{e^{-r_t, T(J)}}{\sqrt{v}}\Phi'\left(\frac{x-\tilde{\alpha}}{\sqrt{v}}\right)\right\}}\left(\frac{x-\tilde{\alpha}}{v}\right)\right\} > E_t\left\{\frac{\frac{1}{\sqrt{v}}\Phi'\left(\frac{x-\alpha}{\sqrt{v}}\right)}{\tilde{E}_t\left\{\frac{1}{\sqrt{v}}\Phi'\left(\frac{x-\alpha}{\sqrt{v}}\right)\right\}}\left(\frac{x-\alpha}{v}\right)\right\}.$$

which simplifies to (B11). ■

The next lemma is used in the proof of part (e) of Proposition 5.

**Lemma B3** *Suppose the random variables  $Y$  and  $Z$  are independent, and  $Z$  is standard normally distributed. Then*

$$\frac{d}{dx}P(Y \leq k | Y + Z = x) \leq 0, \quad \text{for all } k, x \in \mathbb{R}. \quad (\text{B12})$$

*That is, an increase in  $x$  induces a first-order stochastic dominance shift to the right in the conditional distribution. The inequality in (B12) is strict at  $k$  if  $P(Y > k) > 0$  and  $P(Y \leq k) > 0$ .*

**Proof.** Let  $F$  and  $\Phi$  denote the distribution functions of  $Y$  and  $Z$ , respectively. Differentiating

$$P(Y \leq k | Y + Z = x) = \frac{\int_{y \leq k} \Phi'(x - y) dF(y)}{\int \Phi'(x - y) dF(y)},$$

we get  $\frac{d}{dx} P(Y \leq k | Y + Z = x) \leq 0$  if and only if

$$\int \Phi'(x - y) dF(y) \int_{y \leq k} y \Phi'(x - y) dF(y) \leq \int_{y \leq k} \Phi'(x - y) dF(y) \int \Phi'(x - y) y dF(y). \quad (B13)$$

Obviously  $P(Y \leq k) = 0$  implies  $\frac{d}{dx} P(Y \leq k | Y + Z = x) = 0$ . Henceforth assume  $P(Y \leq k) > 0$ . The inequality (B13) is equivalent to

$$E(Y | Y + Z = x, Y \leq k) = \frac{\int_{y \leq k} y \Phi'(k - y) dF(y)}{\int_{y \leq k} \Phi'(x - y) dF(y)} \leq \frac{\int y \Phi'(x - y) dF(y)}{\int \Phi'(x - y) dF(y)} = E(Y | Y + Z = x),$$

which clearly holds. The inequality is strict if both  $P(Y > k)$  and  $P(Y \leq k)$  are strictly positive. ■

## Proof of Lemma 1

Parts (b)-(d) are special cases of the same parts of Proposition 5. The SDF expression in part (a) is a special case of part (e) of Proposition 5. The well-know B-S SDF is

$$m_{BS}^i(x) = \exp \left\{ - (x - \xi^i \tau) \eta^c \rho^i / \sigma^i + \left( \frac{1}{2} (\eta^c \rho^i)^2 - r \right) \tau \right\}.$$

We now prove the claim that  $\delta > 0$  implies nonmonotonicity for sufficiently small  $\tau$ . This is shown by taking the limit of the SDF expression (6) as  $\tau \rightarrow 0$ :

$$e^{r_{t,T}} m_{t,T}(y + \xi \tau) \rightarrow \begin{cases} \exp \left\{ -y \frac{\eta}{\sigma} \right\} & \text{if } \frac{y}{\psi} \leq \frac{1}{2}, \\ \left( \frac{\bar{\lambda}}{\lambda} \right)^j \exp \left( -[y - j\psi] \left( \frac{\eta}{\sigma} \right) \right) & \text{if } -\frac{1}{2} < \frac{y}{\psi} - j \leq \frac{1}{2}, j = 1, 2, \dots \end{cases}.$$

As explained in the discussion following Lemma 1,  $\delta > 0$  implies a positive jump in the limiting SDF at log returns at  $n\psi/2$ ,  $n = 1, 2, \dots$

## Proof of Proposition 5

The risk-neutral expected gross return (recall  $d\tilde{B}_t^c = dB_t^c + \eta_t^c dt$  is standard Brownian motion under  $\tilde{P}$ ) satisfies

$$\tilde{E}(S_1^i/S_0^i) = \exp \left\{ \left( \mu - \sigma^i \rho \eta^c + \ln \left\{ \tilde{E}(J_1^i) \right\} \right) \right\} = \exp(r)$$

for all  $t$ , which implies

$$\mu^i = r + \sigma^i \rho \eta^c - \ln \left\{ \tilde{E}(J_1^i) \right\}.$$

a & b) To derive (16) and (17), we apply Lemma B2 with

$$\tilde{\alpha}(Z) = \xi\tau + \ln(J_T^i/J_t^i), \quad \alpha(Z) = \tilde{\alpha}(Z) + \sigma^i \rho \eta^c \tau, \quad v(Z) = \sigma^i \tau.$$

c) To derive (18), we use (16) to get

$$\begin{aligned} \frac{d}{d(\rho^i \eta^c)} \frac{d}{dy} \ln \{m^i(y + \xi\tau)\} &= -\frac{d}{d(\rho^i \eta^c)} \frac{d}{dy} \ln E_t \left\{ \Phi' \left( \frac{y - \ln(J_T^i/J_t^i) - \sigma^i \rho^i \eta^c \tau}{\sigma^i \sqrt{\tau}} \right) \right\} \\ &= \sigma^i \tau \frac{d^2}{dz^2} \ln E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}, \end{aligned}$$

where we let  $z = y - \sigma^i \rho^i \eta^c \tau$  to simplify notation. Differentiating with respect to  $z$ :

$$\frac{d}{dz} \ln E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\} = \frac{1}{(\sigma^i)^2 \tau} \left( \frac{E_t \left\{ \ln(J_T^i/J_t^i) \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}}{E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}} - z \right).$$

Differentiating again:

$$\begin{aligned}
& (\sigma^i)^2 \tau \frac{d^2}{dz^2} \ln E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\} + 1 \\
&= \frac{1}{(\sigma^i)^2 \tau} \left( \frac{E_t \left\{ [\ln(J_T^i/J_t^i)]^2 \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}}{E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}} - \left[ \frac{E_t \left\{ \ln(J_T^i/J_t^i) \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}}{E_t \left\{ \Phi' \left( \frac{z - \ln(J_T^i/J_t^i)}{\sigma^i \sqrt{\tau}} \right) \right\}} \right]^2 \right) \\
&= \frac{1}{\tau} \text{Var}_t \left\{ \frac{\ln(J_T^i/J_t^i)}{\sigma^i} \middle| z \right\}.
\end{aligned}$$

d) See Lemma B4 below.

e) We omit the superscript  $i$  throughout, except in  $B^i$ . Recall our definition  $m(x) = E(\pi_\tau | \ln(S_\tau/S_0) = x)$ . Defining  $\zeta = \mu - \frac{1}{2}\sigma^2$  then

$$m(x + \zeta\tau) \propto E \left( \exp(-\eta^c B_\tau^c - \theta \ln(J_\tau)) \middle| \sigma^i \left( \rho B_\tau^c + \sqrt{1 - \rho^2} B_\tau^i \right) + \psi \ln(J_\tau) = x \right).$$

Defining the standard Brownian motion  $W_\tau = \rho B_\tau^c + \sqrt{1 - \rho^2} B_\tau^i$ , and introducing a new standard Brownian motion  $\tilde{B}$  assumed independent of  $W$  and  $J$ , then

$$\begin{aligned}
& E \left( \exp(-\eta^c B_\tau^c - \theta \ln(J_\tau)) \middle| \sigma \left( \rho B_\tau^c + \sqrt{1 - \rho^2} B_\tau^i \right) + \psi \ln(J_\tau) \right) \\
&= E \left( \exp \left( -\eta^c \left\{ \rho W_\tau + \sqrt{1 - \rho^2} \tilde{B}_\tau \right\} - \theta \ln(J_\tau) \right) \middle| \sigma W_\tau + \psi \ln(J_\tau) \right) \\
&= E \left\{ \exp \left( -\eta^c \sqrt{1 - \rho^2} \tilde{B}_\tau \right) \right\} E \left\{ \exp(-\eta^c \rho W_\tau - \theta \ln(J_\tau)) \middle| \sigma W_\tau + \psi \ln(J_\tau) \right\}
\end{aligned}$$

Therefore

$$m(x + \zeta\tau) \propto E \left( \exp(-\eta^c \rho W_\tau - \theta \ln(J_\tau)) \middle| \sigma W_\tau + \psi \ln(J_\tau) = x \right).$$

Substituting for  $\theta$  from (20):

$$\begin{aligned} m(x + \zeta\tau) &\propto E \left( \exp \left( -\frac{\eta^c \rho}{\sigma} [W_\tau + \psi \ln(J_\tau)] + \delta \psi \ln(J_\tau) \right) \middle| \sigma W_\tau + \psi \ln(J_\tau) = x \right) \\ &= \exp \left( -\frac{\eta^c \rho}{\sigma} x \right) E \left( \exp(\delta \psi \ln(J_\tau)) \middle| \sigma W_\tau + \psi \ln(J_\tau) = x \right). \end{aligned}$$

That is,

$$m(x) \propto m_{BS}(x) E \left( J_\tau^{\delta \psi} \middle| \ln(S_\tau/S_0) = x \right).$$

Lemma B3 implies that  $E \left( J_\tau^{\delta \psi} \middle| \ln(S_\tau/S_0) = x \right)$  is strictly increasing (decreasing) in  $x$  if  $\delta > 0$  ( $\delta < 0$ ).

Part a) of the following lemma is proven in a pure-diffusion setting with stochastic volatility by Ledoit, Santa-Clara, and Yan (2002); and parts a) and b) for a special case our jump-diffusion model (displaced lognormal jumps) by Yan (2011).

**Lemma B4** *Suppose the stock-price model (13) where the Levy process  $J$  is pure-jump with bounded risk-neutral intensity process. Let  $\hat{\sigma}(\cdot)$  denote the B-S implied volatility curve, and fix the "at-the-money" strike price  $K = e^{r\tau} S_0$ .*

*a) The implied volatility at  $K$  satisfies*

$$\hat{\sigma}(K) = \sigma + O(\sqrt{\tau}). \quad (\text{B14})$$

*b) The slope of the implied volatility curve at  $K$  satisfies*

$$\hat{\sigma}'(K) = \frac{\ln \left\{ \tilde{E}(J_1^i) \right\}}{K \sigma^i} + O(\sqrt{\tau}). \quad (\text{B15})$$

**Proof.** Fix throughout the strike price  $K = e^{r\tau} S_0$ .

**Part a)** Because the put payout is bounded and the probability of at least one jump before expiration is of order  $O(\tau)$  we have the following expression for the price  $p$  of a

European put option (note that  $\ln \left\{ \tilde{E}(J_1^i) \right\}$  is analogous to a dividend rate):

$$\begin{aligned} \frac{p}{e^{-r\tau}K} = & -e^{-\ln\{\tilde{E}(J_1^i)\}\tau} \Phi \left( - \left( -\frac{\ln\{\tilde{E}(J_1^i)\}}{\sigma} + \frac{1}{2}\sigma \right) \sqrt{\tau} \right) \\ & + \Phi \left( - \left( -\frac{\ln\{\tilde{E}(J_1^i)\}}{\sigma} - \frac{1}{2}\sigma \right) \sqrt{\tau} \right) + O(\tau). \end{aligned}$$

Using the Taylor series expansions

$$\Phi(y) = \frac{1}{2} + \frac{y}{\sqrt{2\pi}} + O(y^3) \quad (\text{B16})$$

and  $e^x = 1 + x + O(x^2)$  we get

$$\frac{p}{e^{-r\tau}K} = \frac{1}{\sqrt{2\pi}}\sigma\sqrt{\tau} + O(\tau). \quad (\text{B17})$$

The B-S implied volatility  $\hat{\sigma}(K)$  satisfies<sup>55</sup>

$$p(K) = p^{BS}(K, \hat{\sigma}(K)) \quad \text{all } K > 0,$$

where  $p^{BS}$ , the B-S put price, satisfies

$$\frac{p^{BS}(K, \hat{\sigma}(K))}{e^{-r\tau}K} = \Phi\left(\frac{1}{2}\hat{\sigma}(K)\sqrt{\tau}\right) - \Phi\left(-\frac{1}{2}\hat{\sigma}(K)\sqrt{\tau}\right) = \frac{1}{\sqrt{2\pi}}\hat{\sigma}(K)\sqrt{\tau} + O(\tau^{3/2}). \quad (\text{B18})$$

Matching (B17) and (B18) yields (B14).

**Part b)** It is well known that any option pricing formula differentiable with respect to strike satisfies, assuming a constant interest rate  $r$ ,

$$-e^{r\tau} \frac{\partial c(K)}{\partial K} = \tilde{P}(S_\tau \geq K). \quad (\text{B19})$$

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<sup>55</sup>It is well known that put-call parity implies that the implied volatilities for European calls and puts are identical.

The B-S implied volatility  $\hat{\sigma}(K)$  satisfies

$$c(K) = c^{BS}(K, \hat{\sigma}(K)) \quad \text{all } K > 0, \quad (\text{B20})$$

where  $c^{BS}$  is the B-S put price. Following the approach of Yan (2011), we totally differentiate (B20) to get

$$\begin{aligned} \frac{\partial}{\partial K} c(K) &= c_K^{BS}(K, \hat{\sigma}(K)) + c_\sigma^{BS}(K, \hat{\sigma}(K)) \hat{\sigma}'(K) \\ &= -e^{-r\tau} \Phi\left(-\frac{1}{2} \hat{\sigma}(K) \sqrt{\tau}\right) + e^{-r\tau} K \Phi'\left(-\frac{1}{2} \hat{\sigma}(K) \sqrt{\tau}\right) \sqrt{\tau} \hat{\sigma}'(K). \end{aligned}$$

Using the approximations (B16) and  $\Phi'(y) = \frac{1}{\sqrt{2\pi}} \{1 - \frac{1}{2}y^2\} + O(y^4)$ , and substituting (B19) yields

$$\tilde{P}(S_\tau \geq K) = \frac{1}{2} - \frac{1}{2\sqrt{2\pi}} \hat{\sigma}(K) \sqrt{\tau} - K \frac{1}{\sqrt{2\pi}} \sqrt{\tau} \hat{\sigma}'(K) + O(\tau^{3/2}) \quad (\text{B21})$$

For our stock-price model (13) (with bounded risk-neutral intensity process) the probability of at least one jump before expiration is of order  $O(\tau)$ , and therefore

$$\begin{aligned} \tilde{P}(S_\tau > K) &= \tilde{P}\left(-\left[\ln\left\{\tilde{E}(J_1^i)\right\} + \frac{1}{2}(\sigma^i)^2\right]\tau + \sigma^i\left(\rho\tilde{B}_\tau^c + \sqrt{1-\rho^2}\tilde{B}_\tau^i\right) > 0\right) + O(\tau) \\ &= \Phi\left(-\left\{\frac{\ln\left\{\tilde{E}(J_1^i)\right\}}{\sigma^i} + \frac{1}{2}\sigma^i\right\}\sqrt{\tau}\right) + O(\tau) \\ &= \frac{1}{2} - \frac{1}{\sqrt{2\pi}} \left\{\frac{\ln\left\{\tilde{E}(J_1^i)\right\}}{\sigma^i} + \frac{1}{2}\sigma^i\right\} \sqrt{\tau} + O(\tau). \end{aligned} \quad (\text{B22})$$

Matching the expressions in (B21) and (B22) and using (B14) we get (B15). ■

## Proof of Equation (15)

In the Levy case with absolutely continuous compensator, the (strictly positive) jump process satisfies

$$\frac{dJ_t^i}{J_{t-}^i} = \left( \ln \tilde{E} J_1^i \right) dt + \frac{dM_t^i}{M_{t-}^i}, \quad i = 1, \dots, n,$$

for some strictly positive martingale, which has the solution

$$J_t^i = \exp \left\{ \left( \ln \tilde{E} J_1^i \right) t \right\} M_t^i = \left( \tilde{E} J_1^i \right)^t M_t^i.$$

Then the stock index satisfies

$$\frac{dJ_t^\phi}{J_{t-}^\phi} = \sum_{i=1}^n \phi^i \left( \ln \tilde{E} J_1^i \right) dt + \sum_{i=1}^n \phi^i \frac{dM_t^i}{M_{t-}^i}$$

which has the solution

$$J_t^\phi = \exp \left\{ \left( \ln \tilde{E} J_1^\phi \right) t \right\} M_t^\phi.$$

where

$$\ln \tilde{E} J_1^\phi = \sum_{i=1}^n \phi^i \left( \ln \tilde{E} J_1^i \right)$$

and  $M^\phi$  satisfies

$$\frac{dM_t^\phi}{M_{t-}^\phi} = \sum_{i=1}^n \phi^i \frac{dM_t^i}{M_{t-}^i}.$$

## Appendix C Skewness-adjusted t-statistics

The proposed statistic for testing the mean of positively skewed distributions is taken from Chen (1995) and is derived using the Edgeworth expansion as follows:

$$P \left\{ \frac{\sqrt{n}(\bar{X} - \mu)}{S} \leq x - \frac{1}{6\sqrt{n}} \hat{\beta}_1 (1 + 2x^2) \right\} = \Phi(x) + o(n^{-1/2}), \quad (\text{C1})$$

where  $\Phi(x)$  is the standard normal distribution,  $S^2 = \frac{\sum (X_i - \bar{X})^2}{(n-1)}$ , and  $\hat{\beta}_1 = \frac{n^{-1} \sum (X_i - \bar{X})^3}{S^3}$  (For more details on the Edgeworth expansion, see Hall (1983)). To test the hypothesis

$H_0 : \mu_x = \mu_0$  against  $H_0 : \mu_x > \mu_0$  is to reject  $H_0$  when

$$\frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > z_\alpha - \frac{1}{6\sqrt{n}}\hat{\beta}_1(1 + 2z^2), \quad (\text{C2})$$

where  $z_\alpha$  satisfies the equation  $1 - \Phi(z_\alpha) = \alpha$ . Chen (1995) argues that the critical point of the above hypothesis test depends on the skewness of each sample. Thus to find a more accurate and powerful test with a common critical point for a given significance level  $\alpha$ , we use the following approach. We first solve  $\frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > x - \frac{1}{6\sqrt{n}}\hat{\beta}_1(1 + 2x^2)$  for  $x$ . Let  $\alpha = \frac{\hat{\beta}_1}{6\sqrt{n}}$  and  $t = \frac{\sqrt{n}(\bar{X} - \mu)}{S}$ . Thus, when  $n$  is large such that  $1 - 8a(t + a) \geq 0$ , we have from equation C1,

$$P \left\{ \frac{1 - \sqrt{1 - 8a(t + a)}}{4a} > z_\alpha \text{ or } \frac{1 + \sqrt{1 - 8a(t + a)}}{4a} < z_\alpha \right\} = \alpha + o(n^{-1/2}),$$

or

$$P \left\{ \frac{1 - \sqrt{1 - 8a(t + a)}}{4a} > z_\alpha \right\} \leq \alpha.$$

Using Taylor series expansion we then have

$$\frac{1 - \sqrt{1 - 8a(t + a)}}{4a} = t + a + 2at^2 + 4a^2(t + 2t^3) + o(n^{-1/2}). \quad (\text{C3})$$

Equation C3 is the skewness adjusted t-statistic that we have used in this paper. An advantage of this test statistic is that it can be used for sample sizes as small as 13 and for distributions as asymmetric as the exponential distribution. Expressing equation C3 in terms of the original variables, we get the following expression:

$$t_{skew} = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} + \frac{1}{6\sqrt{n}}\hat{\beta}_1 \left[ 1 + 2 \left( \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right)^2 \right] + \frac{1}{9n}\hat{\beta}_1^2 \left[ \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} + 2 \left( \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right)^2 \right]. \quad (\text{C4})$$

## Appendix D Joint Monotonicity Tests

### D1 Modified Page Test

The Page test for ordered alternatives tests whether average returns are increasing in strike by measuring the significance of the covariance between the rank of average returns and the strike-group number.<sup>56</sup> This covariance will tend to be positive (negative) if higher strikes are associated with higher (lower) ranks. Let  $(X_t^1, \dots, X_t^k)$  (which is some permutation of  $\{1, \dots, k\}$ ) denote the vector of ranks of the time- $t$  average returns for each strike group,<sup>57</sup> and let  $L_t = \frac{1}{k} \sum_{j=1}^k (j - \frac{k+1}{2}) X_t^j$  denote the time- $t$  sample covariance between the ranks and the strike-group number. The Page statistic is the time-series average of these covariances divided by the corresponding standard deviation.<sup>58</sup> The classical Page test is based on the (typically unstated) assumption that for each sample drawing, each of the  $k!$  possible ranks are equally likely under the null.<sup>59</sup> In our application this assumption is questionable because, for example, in the Black-Scholes model under the null (which is equivalent to zero risk premia) only two ranks of realized call-option returns would be observed for a particular stock over some small period: returns monotonically increasing in strike if the market return exceeds the riskless return, and returns monotonically decreasing in strike otherwise. This tends to increase the variance of the test statistic, though the effect is mitigated by both averaging of returns across stocks each period and by other factors (such as volatility) affecting call-option returns. Rather than assuming that each permutation is equally likely, the null hypothesis of our modified test statistic is no linear relationship

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<sup>56</sup>Although, our focus is on testing monotonically increasing alternatives, this test, in principle, can be modified to test other patterned alternatives (for example, U-shaped kernels).

<sup>57</sup>We take the average return differences from our pairwise tests (Section IVA) and, for each period, cumulate them across strike groups to get an average return for each group. Strike group zero is assigned a zero average return; average returns for all other groups are relative to strike group zero.

<sup>58</sup>Unlike the Spearman rank correlation test, the Page test allows dependence among each time- $t$  set of observations.

<sup>59</sup>Equally likely ranks implies

$$\text{Var}(L_t) = \frac{1}{144} (k+1)^2 (k-1),$$

which yields the classical Page statistic.

between rank and strike-group number:  $E(L_t) = 0$  for all  $t$ .<sup>60</sup>

Our modified Page test statistic,  $\hat{P}$ , is

$$\hat{P} = \frac{1}{\hat{\sigma}_B \sqrt{T}} \sum_{t=1}^T L_t, \quad \hat{\sigma}_P^2 = \frac{1}{T} \sum_{t=1}^T L_t^2. \quad (D1)$$

If the vector of ranks  $(X_t^1, \dots, X_t^k)$  is i.i.d. over time, then  $\hat{P}$  is asymptotically standard normal under the null.<sup>61</sup>

## D2 Bootstrapping Test

Patton and Timmermann (2010) propose a general test for monotonicity using bootstrapping. They specify an alternative hypothesis that the minimum increment in expected returns across strikes is strictly positive (that is, expected-returns are strictly increasing in strike) and a null that the minimum increment is nonpositive. The test statistic in our application is the minimum increment across strikes in time-series average returns. If the test statistic is sufficiently large relative to its empirical distribution estimated under the null via bootstrapping, then the null is rejected in favor of monotonicity. Their procedure is applied in our setting as follows:

- i) Compute the vector of time-series average returns by strike,  $(\hat{\mu}_1, \dots, \hat{\mu}_k)$ , and define  $\hat{\Delta} = (\hat{\Delta}_1, \dots, \hat{\Delta}_{k-1})'$  as the vector of differences:  $\hat{\Delta}_i = \hat{\mu}_{i+1} - \hat{\mu}_i$ . Let  $J =$

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<sup>60</sup>A special case is if each permutation is equally likely (as in the classical Page test). Another special case is if the distribution of the rank is symmetric under  $H_0$  in the sense that the distribution of  $(X_t^1, \dots, X_t^k)$  is the same as the distribution  $(X_t^k, \dots, X_t^1)$ , which includes the case of only two equally likely rankings: strictly ascending or strictly descending.

<sup>61</sup>Closely related to Page is the test of Bathke (2009), which is (D1) but with  $X$  replaced by  $Y$  in the definition of  $L_t$ , where  $(Y_1^1, \dots, Y_T^k)$  denotes the length- $T \cdot k$  vector of ranks of the entire sample of average returns across time and strikes (which is some permutation of  $\{1, \dots, T \cdot k\}$ ). That is, the Page test is based on the per-period ranks of average returns whereas the Bathke (2009) test is based on the overall average ranks. The assumptions for asymptotic standard normality of  $\hat{B}$  under the null are therefore stronger. In particular, Bathke (2009) assumes that the average return vectors are i.i.d. over time. Heteroscedasticity is likely to be more of a problem with the Bathke (2009) tests because, for example, increasing variance over time is likely to result in more extreme ranks appearing later in the sample, giving more weight to those observations.

$\min_{\{i=1,\dots,k-1\}} \hat{\Delta}_i$  denote the the minimum incremental average return for the entire sample.

ii) We compute the empirical distribution of the test statistic  $J$  under the null using bootstrapping.

(a) The bootstrap samples are formed using the stationary bootstrap method introduced in Politis and Romano (1994) and used in Patton and Timmermann (2010). As in any bootstrapping procedure, the stationary bootstrap re-samples the original data repeatedly to estimate the distribution of the statistic.<sup>62</sup> Unlike traditional bootstrap procedures, the stationary bootstrap retains the time series properties of the original sample by sampling returns data in random blocks, where the length of each block is drawn from a geometric distribution.

(b) For each bootstrap subsample  $b$ , we compute the vector  $\hat{\Delta}^{(b)}$  of increments in the average returns of the subsample (analogous to step 1).

(c) The null hypothesis least favorable to the alternative,  $\mu_1 = \dots = \mu_k$ , is imposed by subtracting  $\hat{\Delta}$  from  $\hat{\Delta}_i^{(b)}$ . The bootstrapped value of the statistic under the null is then  $J^{(b)} = \min_{\{i=1,\dots,k-1\}} \left( \hat{\Delta}_i^{(b)} - \hat{\Delta}_i \right)$ .

iii) From the empirical distribution of  $J^{(b)}$  we find the  $p$ -value corresponding to  $\hat{\Delta}$ :

$$p = \frac{1}{B} \sum_{b=1}^B 1 \{ J^{(b)} > J \},$$

where  $B$  denotes the number of bootstrap samples (1,000 in our application). That is,  $p$  represents the proportion of bootstrapped estimates  $J^{(b)}$  that exceed  $J$ . For sufficiently small  $p$  (i.e., sufficiently large  $J$ ) we reject the null in favor of strict monotonicity.

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<sup>62</sup>We thank Andrew Patton for providing the MATLAB code for the stationary bootstrap on his website.

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**Table I**  
**Descriptive statistics of strikes and moneyness**

This table presents descriptive statistics of number of strikes and moneyness for option contracts in the sample. Panel A reports the number of stocks that have “x” number of strikes. “No. of stock-weeks” is the number of “stock x week” combinations that have “x” number of strikes and “No. of unique stocks” is the number of stocks that have “x” number of strikes where “x” is now the median number of strikes for each underlying stock for the entire sample. Panel B reports the mean and median value of moneyness for each strike group. Moneyness is defined as the ratio of strike price and the current underlying stock price.

| Table I: Descriptive statistics    |        |        |         |         |           |          |
|------------------------------------|--------|--------|---------|---------|-----------|----------|
| Panel A: Distribution of strikes   |        |        |         |         |           |          |
| No. of strikes                     | 1      | 2      | 3       | 4       | $\geq 5$  | $\geq 7$ |
| No. of stock-weeks                 | 13,700 | 45,893 | 544,901 | 297,394 | 1,169,623 | 729,486  |
| No. of unique stocks               | 16     | 69     | 2,073   | 1,670   | 3,868     | 1,874    |
| Panel B: Moneyness by strike group |        |        |         |         |           |          |
| Strike Group                       | 1      | 2      | 3       | 4       | 5         | $> 5$    |
| Mean moneyness                     | 0.768  | 0.852  | 1.029   | 1.241   | 1.474     | 1.641    |
| Median moneyness                   | 0.786  | 0.878  | 1.001   | 1.136   | 1.277     | 1.398    |

**Table II**  
**Return differences for option strategies on individual stocks**

This table reports differences of weekly average buy and hold returns between January 1996 and December 2012 for some option trading strategies that are special cases of the log-concave class of payout functions. All options have between 6 weeks and 3 weeks to maturity and returns are calculated using prices observed on Tuesdays. The options must satisfy the following conditions to be included in the sample: (1) the ask price is greater than the bid price, (2) the underlying stock does not have an ex-dividend date prior to maturity and (3) the option prices satisfies a no-arbitrage restriction.  $R_s$  denotes the return on the underlying stock and  $R_f$  denotes the risk-free rate.

| Table II: Average return differences by strike group |         |       |       |        |           |
|------------------------------------------------------|---------|-------|-------|--------|-----------|
| Panel A: call returns                                |         |       |       |        |           |
| Strike Groups                                        | $1-R_s$ | 2 -1  | 3 -2  | 4 -3   | 5 -4      |
| Average returns                                      | 0.002   | 0.000 | 0.012 | 0.067  | 0.026     |
| T-statistics                                         | 0.485   | 0.199 | 3.625 | 25.983 | 6.737     |
| Skewness adjusted t-statistics                       | 0.488   | 0.197 | 3.841 | 43.251 | 6.338     |
| Panel B: put returns                                 |         |       |       |        |           |
| Strike Groups                                        | 2 -1    | 3 -2  | 4 -3  | 5 -4   | $R_f - 5$ |
| Average returns                                      | 0.004   | 0.009 | 0.003 | 0.001  | 0.004     |
| T-statistics                                         | 2.313   | 2.423 | 1.736 | 1.264  | 1.048     |
| Skewness adjusted t-statistics                       | 2.315   | 2.302 | 1.698 | 1.250  | 1.039     |
| Panel C: butterfly spread returns                    |         |       |       |        |           |
| Strike Groups                                        | 2 -1    | 3 -2  | 4 -3  | 5 -4   |           |
| Average returns                                      | -       | 0.026 | 0.049 | 0.028  | 0.067     |
| T-statistics                                         | -       | 8.422 | 8.012 | 4.472  | 9.422     |
| Skewness adjusted t-statistics                       | -       | 7.781 | 7.520 | 4.557  | 10.671    |

**Table III**  
**Joint tests of monotonicity for individual stocks**

This table presents two joint tests of monotonicity for individual stocks. We use the Page test for ordered alternatives to test whether average strategy returns tend to be increasing in the strike price. We report the statistics for the classical Page test and our modified Page test statistic that does not depend on the equal rank assumption under the null. We also report the results from a general test of monotonicity using bootstrapping introduced in Patton and Timmermann (2010). The test statistic is the minimum increment across strikes in time-series average returns. The null hypothesis is that the minimum increment in average strategy returns is nonpositive. If the test statistic is sufficiently large relative to its empirical distribution estimated under the null via bootstrapping, then the null is rejected in favor of monotonicity. In this table, we report the p-values as well as studentized p-values from the bootstrapping test.

| Table III: Joint tests of monotonicity |                |                     |          |                             |             |
|----------------------------------------|----------------|---------------------|----------|-----------------------------|-------------|
|                                        | No. of Periods | Page Test Statistic |          | Bootstrapping Test p-values |             |
|                                        |                | Classical           | Modified | Non-Studentized             | Studentized |
| call                                   | 807            | 22.571              | 19.523   | 0.098                       | 0.092       |
| put                                    | 807            | 15.358              | 9.192    | 0.013                       | 0.004       |
| butterfly spread                       | 807            | 14.869              | 8.36     | 0.003                       | 0.001       |

**Table IV**  
**Return differences for option strategies on the S&P 500 index**

This table reports differences of weekly average buy and hold returns between January 1996 and December 2012 for some option trading strategies on the S&P 500 index that are special cases of the log-concave class of payout functions. All options have between 6 weeks and 3 weeks to maturity and returns are calculated using prices observed on Tuesdays. The options must satisfy the following conditions to be included in the sample: (1) the ask price is greater than the bid price, and (2) the option prices satisfies a no-arbitrage restriction.  $R_s$  denotes the return on the S&P 500 index and  $R_f$  denotes the risk-free rate.

| Table IV: Average return differences by strike group |          |        |        |       |           |
|------------------------------------------------------|----------|--------|--------|-------|-----------|
| Panel A: call returns                                |          |        |        |       |           |
| Strike Groups                                        | 1- $R_s$ | 2 -1   | 3 -2   | 4 -3  | 5 -4      |
| Average returns                                      | -0.033   | -0.003 | -0.001 | 0.004 | 0.001     |
| t statistics                                         | -1.815   | -2.135 | -0.578 | 0.964 | 0.148     |
| Skewness adjusted t statistics                       | -1.687   | -1.859 | -0.555 | 1.167 | 0.0.170   |
| Panel B: put returns                                 |          |        |        |       |           |
| Strike Groups                                        | 2 -1     | 3 -2   | 4 -3   | 5 -4  | $R_f$ - 5 |
| Average returns                                      | 0.008    | 0.005  | 0.004  | 0.006 | 0.020     |
| t statistics                                         | 5.693    | 2.970  | 2.008  | 4.098 | 1.017     |
| Skewness adjusted t statistics                       | 4.982    | 2.731  | 1.788  | 3.617 | 0.988     |
| Panel C: butterfly spread returns                    |          |        |        |       |           |
| Strike Groups                                        | 2 -1     | 3 -2   | 4 -3   | 5 -4  |           |
| Average returns                                      | -        | -0.119 | -0.149 | 0.080 | 0.018     |
| t statistics                                         | -        | -1.067 | -1.556 | 0.848 | 0.175     |
| Skewness adjusted t statistics                       | -        | -1.108 | -1.691 | 0.866 | 0.182     |

**Table V**  
**At-the-Money Implied Volatility Slopes**

This table reports mean and median values of normalized At-the-Money (ATM) implied volatility (IV) slopes and  $\Pi$  for individual stocks sorted by quintiles of normalized ATM IV slopes. Normalized ATM IV slopes are defined as the ATM IV slope times the current underlying stock price.  $\Pi$  is defined as the normalized slope divided by the ATM implied volatility. The last two rows report these values for the full sample and S&P 500 respectively.

|                    | Normalized Slope |        | $\Pi$  |        |
|--------------------|------------------|--------|--------|--------|
|                    | Mean             | Median | Mean   | Median |
| Bottom Quintile    | -0.721           | -0.598 | -2.393 | -1.565 |
| Quintile 2         | -0.337           | -0.323 | -0.936 | -0.815 |
| Quintile 3         | -0.193           | -0.191 | -0.516 | -0.468 |
| Quintile 4         | -0.044           | -0.050 | -0.109 | -0.118 |
| Top Quintile       | 0.389            | 0.270  | 1.481  | 0.655  |
| Full Sample        | -0.181           | -0.189 | -0.495 | 0.418  |
| <b>S&amp;P 500</b> | -0.649           | -0.652 | -3.939 | -3.625 |

**Table VI**  
**Call return differences by steepness of Implied Volatility Curve**

This table reports differences of weekly average buy and hold returns between January 1996 and December 2009 for call options sorted by steepness of the implied volatility (IV) curve of the underlying stock. All options have between 6 and 3 weeks to maturity and returns are calculated using prices observed on Tuesdays. The options must satisfy the following conditions to be included in the sample: (1) the ask price is greater than the bid price, (2) the underlying stock does not have an ex-dividend date prior to maturity and (3) the option prices satisfies a no-arbitrage restriction.  $R_s$  denotes the return on the underlying stock.

| Table VI: Return differences by steepness of IV Curve |          |        |       |        |        |
|-------------------------------------------------------|----------|--------|-------|--------|--------|
|                                                       | 1- $R_s$ | 2 -1   | 3 -2  | 4 -3   | 5 -4   |
| <u>Steepness of IV curve</u>                          |          |        |       |        |        |
| Bottom Quintile                                       | -0.004   | -0.009 | 0.027 | 0.090  | 0.026  |
| t statistic                                           | -1.163   | -5.518 | 7.134 | 25.871 | 5.506  |
| Skewness adjusted t statistics                        | -1.165   | -5.396 | 8.144 | 41.009 | 5.321  |
|                                                       |          |        |       |        |        |
| Quintile 2                                            | 0.000    | -0.002 | 0.004 | 0.071  | 0.035  |
| t statistic                                           | 0.109    | -1.431 | 1.115 | 21.959 | 7.969  |
| Skewness adjusted t statistics                        | 0.109    | -1.421 | 1.144 | 42.279 | 7.589  |
|                                                       |          |        |       |        |        |
| Quintile 3                                            | 0.002    | 0.000  | 0.001 | 0.064  | 0.036  |
| t statistic                                           | 0.460    | 0.039  | 0.296 | 19.119 | 7.683  |
| Skewness adjusted t statistics                        | 0.462    | 0.037  | 0.304 | 38.799 | 7.219  |
|                                                       |          |        |       |        |        |
| Quintile 4                                            | 0.003    | 0.003  | 0.002 | 0.059  | 0.033  |
| t statistic                                           | 0.796    | 1.446  | 0.669 | 19.321 | 6.714  |
| Skewness adjusted t statistics                        | 0.794    | 1.458  | 0.684 | 33.645 | 6.192  |
|                                                       |          |        |       |        |        |
| Top Quintile                                          | 0.006    | 0.008  | 0.009 | 0.043  | 0.055  |
| t statistic                                           | 2.223    | 4.386  | 2.821 | 11.493 | 10.791 |
| Skewness adjusted t statistics                        | 2.211    | 4.438  | 2.924 | 14.819 | 9.442  |

**Table VII**  
**Put return differences by steepness of Implied Volatility Curve**

This table reports differences of weekly average buy and hold returns between January 1996 and December 2009 for put options sorted by steepness of the implied volatility (IV) curve of the underlying stock. All options have between 6 and 3 weeks to maturity and returns are calculated using prices observed on Tuesdays. The options must satisfy the following conditions to be included in the sample: (1) the ask price is greater than the bid price, (2) the underlying stock does not have an ex-dividend date prior to maturity and (3) the option prices satisfies a no-arbitrage restriction.  $R_s$  denotes the return on the underlying stock.

| Table VII: Return differences by steepness of IV Curve |        |       |       |       |           |
|--------------------------------------------------------|--------|-------|-------|-------|-----------|
|                                                        | 2 -1   | 3 -2  | 4 -3  | 5 -4  | $R_f$ - 5 |
| <u>Steepness of IV curve</u>                           |        |       |       |       |           |
| Bottom Quintile                                        | 0.024  | 0.011 | 0.001 | 0.000 | 0.000     |
| t statistic                                            | 14.895 | 3.216 | 0.561 | 0.319 | 0.057     |
| Skewness adjusted t statistics                         | 14.183 | 3.009 | 0.551 | 0.314 | 0.054     |
| Quintile 2                                             | 0.022  | 0.015 | 0.004 | 0.001 | 0.003     |
| t statistic                                            | 14.693 | 4.252 | 1.814 | 1.015 | 0.778     |
| Skewness adjusted t statistics                         | 14.105 | 3.934 | 1.774 | 1.006 | 0.774     |
| Quintile 3                                             | 0.013  | 0.014 | 0.005 | 0.002 | 0.005     |
| t statistic                                            | 7.876  | 3.689 | 1.993 | 1.327 | 1.355     |
| Skewness adjusted t statistics                         | 8.761  | 3.401 | 1.939 | 1.313 | 1.335     |
| Quintile 4                                             | 0.000  | 0.013 | 0.005 | 0.002 | 0.004     |
| t statistic                                            | 0.389  | 3.313 | 2.263 | 1.353 | 1.031     |
| Skewness adjusted t statistics                         | 0.392  | 3.114 | 2.206 | 1.338 | 1.023     |
| Top Quintile                                           | -0.012 | 0.002 | 0.004 | 0.001 | 0.004     |
| t statistic                                            | -3.800 | 0.421 | 1.739 | 1.147 | 1.331     |
| Skewness adjusted t statistics                         | -3.717 | 0.406 | 1.701 | 1.135 | 1.320     |

**Table VIII**  
**Butterfly return differences by steepness of Implied Volatility Curve**

This table reports differences of weekly average buy and hold returns between January 1996 and December 2009 for butterfly call spreads sorted by steepness of the implied volatility (IV) curve of the underlying stock. All options have between 6 and 3 weeks to maturity and returns are calculated using prices observed on Tuesdays. The options must satisfy the following conditions to be included in the sample: (1) the ask price is greater than the bid price, (2) the underlying stock does not have an ex-dividend date prior to maturity and (3) the option prices satisfies a no-arbitrage restriction.

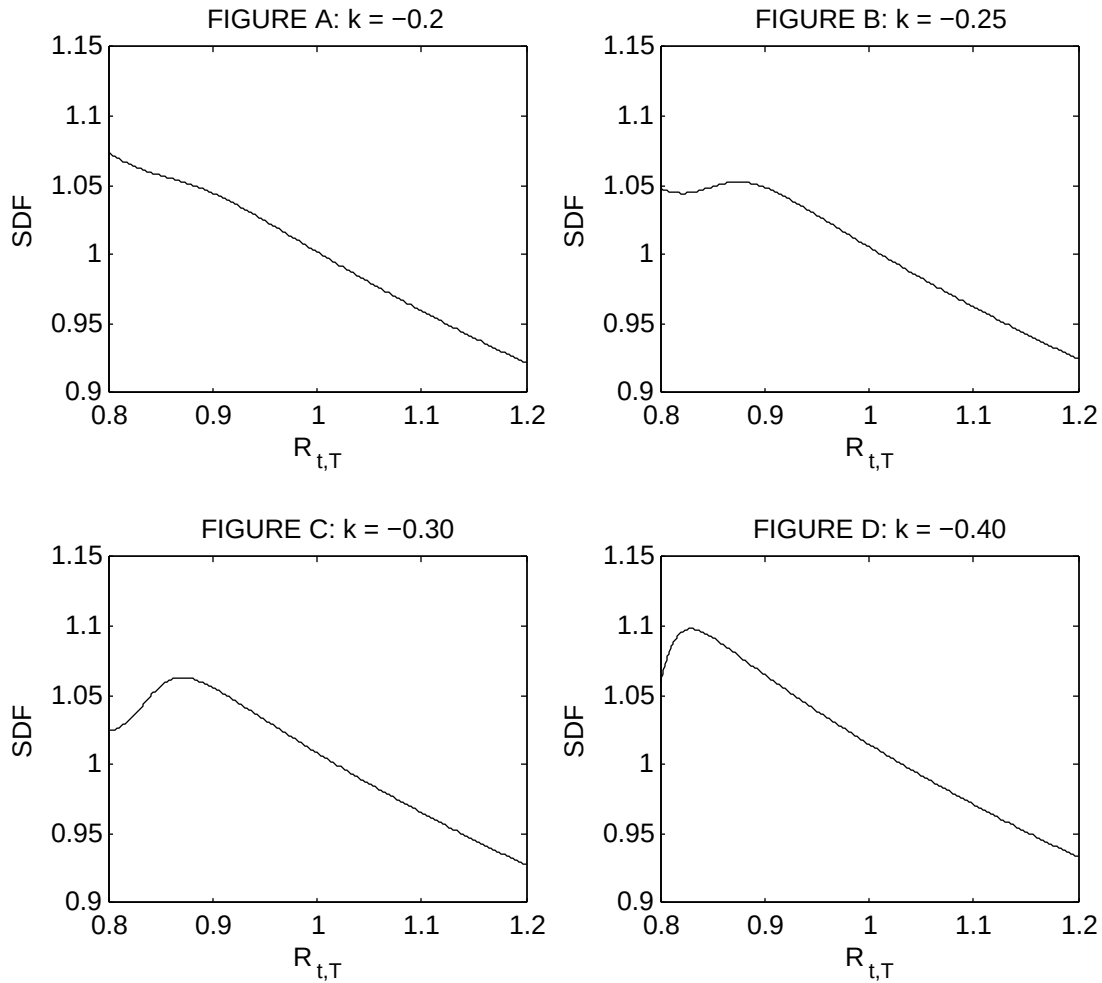
| Table VIII: Return differences by steepness of IV Curve |        |        |        |       |
|---------------------------------------------------------|--------|--------|--------|-------|
|                                                         | 2 -1   | 3 -2   | 4 -3   | 5 -4  |
| Steepness of IV curve                                   |        |        |        |       |
| Bottom Quintile                                         | -0.015 | -0.045 | 0.137  | 0.082 |
| t statistic                                             | -2.099 | -5.797 | 11.542 | 4.339 |
| Skewness adjusted t statistics                          | -2.127 | -5.873 | 16.779 | 5.105 |
| Quintile 2                                              | 0.025  | -0.012 | 0.095  | 0.109 |
| t statistic                                             | 3.992  | -1.813 | 10.575 | 4.731 |
| Skewness adjusted t statistics                          | 3.896  | -1.800 | 12.674 | 8.050 |
| Quintile 3                                              | 0.041  | 0.020  | 0.079  | 0.095 |
| t statistic                                             | 6.693  | 3.187  | 8.566  | 5.749 |
| Skewness adjusted t statistics                          | 6.462  | 3.236  | 8.874  | 6.843 |
| Quintile 4                                              | 0.060  | 0.051  | 0.058  | 0.121 |
| t statistic                                             | 9.611  | 7.327  | 6.109  | 5.359 |
| Skewness adjusted t statistics                          | 9.043  | 7.475  | 6.708  | 6.765 |
| Top Quintile                                            | 0.137  | 0.123  | -0.081 | 0.162 |
| t statistic                                             | 16.909 | 14.164 | -5.589 | 6.399 |
| Skewness adjusted t statistics                          | 19.194 | 17.181 | -6.558 | 6.403 |

**Table IX**  
**Stock characteristics by quintiles of ATM volatility slope**

This table reports mean (median) values of some characteristics of underlying stocks sorted by quintiles of (normalized) At-the-Money (ATM) implied volatility (IV) slope. Normalized ATM IV slopes are defined as the ATM IV slope times the the current underlying stock price. *MV* is the market value of equity. *Market Beta*, *R-Squared*, and *RMSE* are estimated for each underlying stock and each month from the Fama-French and Carhart 4-factor regression model using daily returns. *RMSE (scaled)* is the ratio of the estimated RMSE and the S&P 500 volatility for the same month using daily returns.

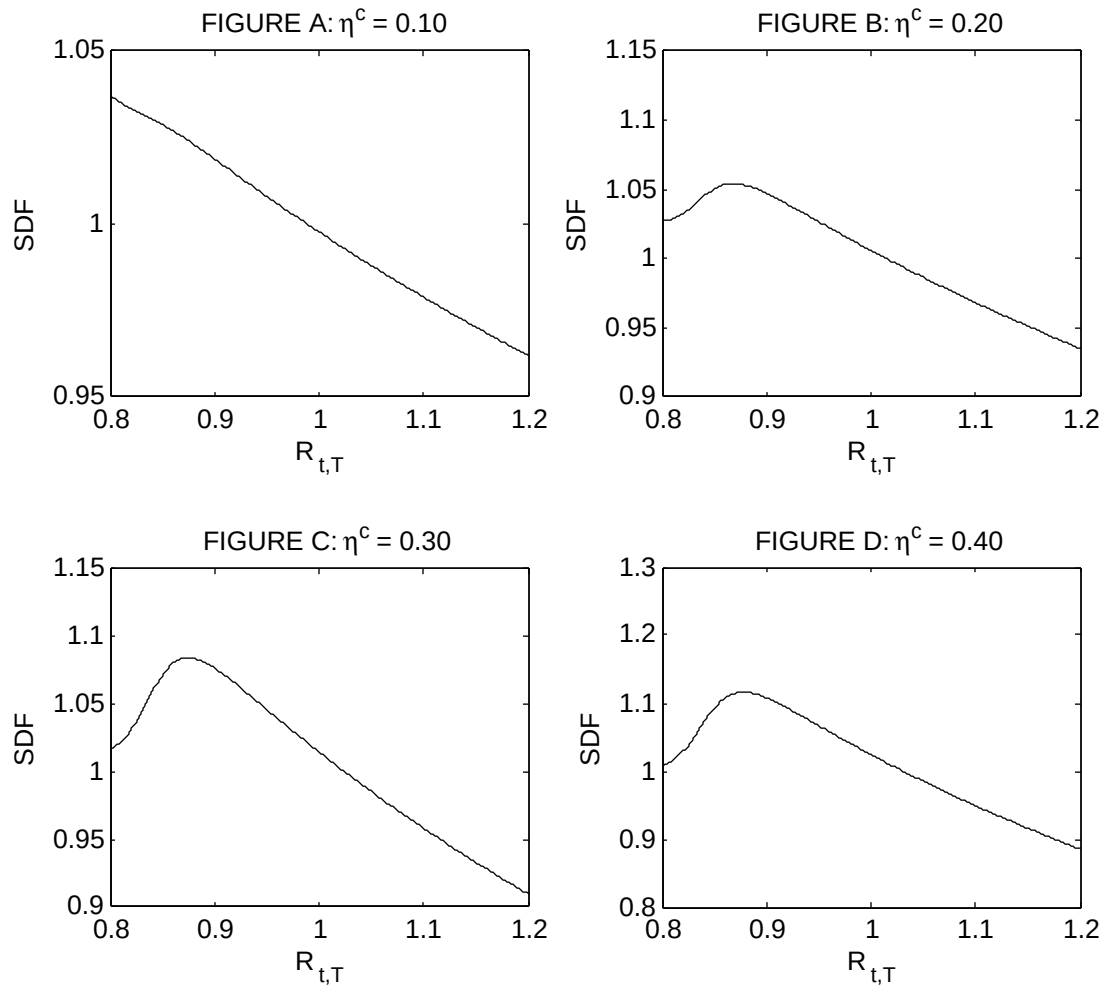
|             | MV(Billions)   | Market Beta   | R- Squared    | RMSE          | RMSE (scaled) |
|-------------|----------------|---------------|---------------|---------------|---------------|
| Q1 (Bottom) | 14.636 (1.563) | 1.070 (0.998) | 0.495 (0.484) | 0.024 (0.017) | 2.493 (1.750) |
| Q2          | 18.492 (1.906) | 1.120 (1.033) | 0.484 (0.481) | 0.024 (0.018) | 2.443 (1.856) |
| Q3          | 15.778 (1.669) | 1.092 (1.042) | 0.472 (0.469) | 0.025 (0.019) | 2.554 (1.950) |
| Q4          | 14.040 (1.307) | 1.074 (1.024) | 0.461 (0.454) | 0.026 (0.020) | 2.675 (2.034) |
| Q5 (Top)    | 9.852 (0.904)  | 0.942 (0.919) | 0.452 (0.439) | 0.028 (0.020) | 2.890 (1.974) |

**Figure 1**  
**Simulated SDFs for the Jump Diffusion Model varying by  $k$**



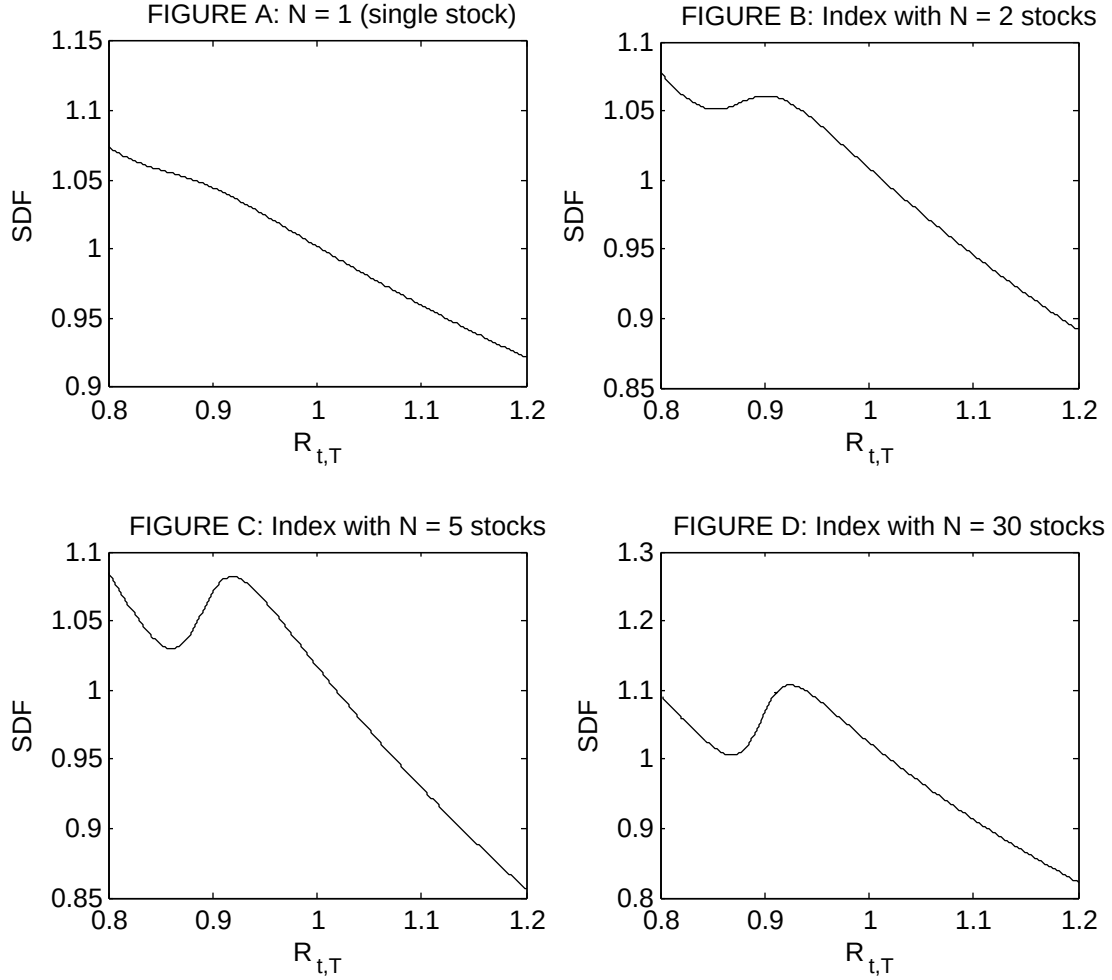
This figure shows plots of SDFs for a single stock estimated as a function of gross returns using expression 6 in Section IIA. Figures A, B, C, and D plots the SDF for the stock for different values of  $k$ .  $R_{t,T}$  is the gross return in the interval  $[t, T]$ . Other parameters used are as follows:  $\lambda = 1.5$ ,  $\bar{\lambda} = 1.6$ ,  $\eta^c = 0.23$ ,  $\rho = 0.6$ ,  $\sigma = 0.3$ ,  $T = 1/12$  years, and  $r = 0.02$

**Figure 2**  
**Simulated SDFs for the Jump Diffusion Model varying by  $\eta^c$**



This figure shows plots of SDFs for a single stock estimated as a function of gross returns using expression 6 in Section IIA. Figures A, B, C, and D plots the SDF for the stock for different values of  $\eta^c$ .  $R_{t,T}$  is the gross return in the interval  $[t, T]$ . Other parameters used are as follows:  $\lambda = 1.5$ ,  $\bar{\lambda} = 1.6$ ,  $k = -0.3$ ,  $\rho = 0.6$ ,  $\sigma = 0.3$ ,  $T = 1/12$  years, and  $r = 0.02$

**Figure 3**  
**Simulated SDFs for the Index varying by  $N$**



This figure shows plots of SDFs for indexes formed by aggregating different number of stocks. The SDF is estimated as a function of gross returns using expression 6 in Section IIA. Figures A, B, C, and D plots the SDF for the index for different values of  $N$ , where  $N = 1$  represents just a single stock. The Index is formed by aggregating stocks with identical parameters.  $R_{t,T}$  is the gross return in the interval  $[t, T]$ . Other parameters used are as follows:  $\lambda = 1.5$ ,  $\tilde{\lambda} = 1.6$ ,  $k = -0.2$ ,  $\rho = 0.6$ ,  $\sigma = 0.3$ ,  $T = 1/12$  years, and  $r = 0.02$